

HEAD AND NECK SURGERY

CHAPTER

1

Thyroid Gland Introduction

APPLIED ANATOMY

Embryology

The thyroid gland originates as a primitive bud from the median pharyngeal wall, at the level which will be in the adult life the junction between the anterior two thirds of the tongue and its posterior third or foramen cecum. It then migrates with the help of the thyroglossal duct (TG) system to its final position; the lower part of the front of the neck. This TG duct then obliterates and disappears.

The fourth branchial arch contributes to the embryology of the thyroid gland through the ultimobranchial body. This is a population of the C-cells migrating out of the neural crest. They are so called because they secrete calcitonin; a calcium lowering agent in the higher primates but not in humans. They are also known as parafollicular cells as they lie in between the thyroid follicles. The neural crest origin of the C-cells sheds some light on the well-known association between the medullary thyroid carcinoma (MTC), arising from the C-cells, and many other tumors, the most popular of which is the multiple endocrine neoplasia (MEN)-II syndrome. Obviously, MTC is *not* thyroid stimulating hormone (TSH)-dependent.

Congenital Anomalies

Such an embryology going wrong, results into a number of anomalies of the thyroid gland and the TG duct system namely:

1. Agenesis of the thyroid gland.
2. Lingual thyroid gland, partly or completely.
3. Median ectopic thyroid tissue.
4. Thyroglossal duct cyst.
5. Combination of numbers 3 and 4.
6. Ectopic parathyroid glands, especially the lower ones.

Morphology

The thyroid gland looks like a butterfly and consists of two lobes, isthmus and a pyramidal lobe. It weighs 25-30 g in average and each lobe measures 5-6 cm long. Sometimes the middle of the thyroid lobe carries a lateral hump well known as *Zuckerkandl* nodule. Its surgical importance is that it usually hides the recurrent laryngeal nerve (RLN) behind it, making its dissection a bit more difficult.

It overrides the trachea, where each lobe extends from the oblique line of the thyroid cartilage above to the sixth tracheal ring below. The isthmus overlies tracheal rings number 2 to 4. Laterally each lobe abuts the carotid sheath.

Arterial Supply

The thyroid gland receives its blood supply from many sources. The first artery that supplies the gland is the superior thyroid artery, a branch of the external carotid artery (ECA). It enters the gland through its superior pole where it divides into at least two branches, the anterior and posterior branches. The posterior one contributes a little to the blood supply of the ipsilateral superior parathyroid gland (PTG). The superior thyroid artery is accompanied by the superior thyroid vein and the external laryngeal nerve.

The second artery is the inferior thyroid artery (ITA), one of the three branches coming out of the thyrocervical trunk, which arises from the first part of the subclavian artery. It enters the gland through its lateral aspect after giving branches to ipsilateral both PTGs representing their main arterial supply. The ITA is closely related to the RLN where the artery is superficial to the nerve in majority of the cases.

Some esophageal and tracheal branches of the corresponding plexuses share in supplying the gland with arterial blood through its deep surface. In a minority of people, the thyroid gland also receives a *thyroidea ima artery* that arises from the aortic arch and enters the gland through the lower border of its isthmus.

Venous Drainage

The venous drainage of the thyroid gland does not match exactly its arterial supply. The upper part of each lobe is drained through the superior thyroid vein, which is a tributary of the internal jugular vein (IJV). It is accompanied by the superior thyroid artery and the external laryngeal nerve.

The lower part of each lobe is drained through a number of veins called inferior thyroid veins that are tributaries of the left innominate vein. The inferior thyroid veins may be closely related to the RLN through its ascent towards the neck.

Some esophageal and tracheal branches of the corresponding plexuses share in draining the gland through its deep surface. In a minority of people, the thyroid gland also has a *thyroidea ima vein* that arises from the lower border of its isthmus and enters the left innominate vein.

Lymphatic Drainage

The thyroid gland is drained through a complex network of lymphatics that eventually carry the lymphatic waste to the cervical lymph nodes (LNs). Understanding the novel work of the Memorial Sloan Kettering Cancer Center (MSKCC) of New York as regards the distribution of the cervical LNs into specific numbered zones, is crucial to the proper management of malignancies, not only of the thyroid gland, but of all tumors of the head and neck region.

In this classification, the neck nodes were distributed into 6 areas in addition to a seventh superior mediastinal area. They are universally known as areas I to VII. Like any other organ, the thyroid gland, drains its lymph in a specific order. The first echelon LNs are present in areas VI and VII, which then drain into areas III, IV, V, II and lastly I. This order is *not* obeyed all the time, as the location of the tumor in the thyroid lobe affects the order of LNs that become affected by malignant cells.

Nerve Supply

Parasympathetic nerve supply comes from the vagus nerve and the sympathetic one comes from the superior, middle and inferior sympathetic ganglia of the sympathetic trunk. These small nerves enter the gland along blood vessels.

Fasciae, Muscles and Nerves

The thyroid gland is encased within fasciae, protected by muscles and closely related to important nerves. In contrast to other body regions, the neck is characterized by having three deep fasciae. These are the investing layer of the deep fascia, the pre-tracheal fascia and the pre-vertebral fascia.

The pre-tracheal fascia is the one that directly envelopes the thyroid gland. It stretches between the thyroid cartilage above and the aortic arch below. Being contained within it, the thyroid gland moves with deglutition. It is also a helping factor in retro-sternal extension of large goiters, as it guides the gland towards the superior mediastinum being attached to the aortic arch. The postero-medial portion of it is very tough and anchors the corresponding part of the thyroid lobe into the trachea. It is called *Berry's ligament*. It hides the RLN behind it just before the entry point of the latter into the larynx. The Berry's ligament carries a small artery and vein within it and therefore should be carefully controlled during total thyroidectomy to avoid bleeding in an area which is extremely close to the genu of the RLN.

Muscles related to the thyroid gland are the strap muscles. These are four muscles stratified into two strata, two superficial and two deep. As a matter of fact, it is a bit difficult to remember their exact order. But two simple rules can make this job much easier. The first is that the longer two are superficial and the shorter two are deep. The second is that the neck structures are always named from down upwards. Applying these two simple rules, make the four muscles easily remembered. These are omohyoid, sternohyoid, sternothyroid and thyrohyoid. The sternothyroid muscle is the direct cover sheet for the thyroid lobe and is the one which is sometimes cut to deliver a huge lobe or excised due its direct malignant invasion by a thyroid neoplasm.

The strap muscles are also called the extrinsic laryngeal muscles because they move the larynx. They are all supplied by the *ansa cervicalis nerve*, which enters them low in the neck. The intrinsic laryngeal muscles on the other hand are those which move the vocal folds. They all lie inside the larynx and are supplied by the RLN except the cricothyroid muscle. It is the only intrinsic laryngeal muscle that lies outside the larynx and is supplied by the external laryngeal nerve. It is the tensor of the vocal folds, while all the other intrinsic laryngeal muscles are responsible for their adduction and abduction.

The nerves that have already been mentioned are those which concern the thyroid surgeon. The external laryngeal nerve is a branch of the superior laryngeal nerve, which is in turn a branch of the vagus nerve. It is a motor nerve that supplies the cricothyroid muscle; the vocal folds' tensor. It is located in the *Joll's triangle*. Its injury usually occurs while ligating the superior thyroid vessels. Its injury results in paralysis of the ipsilateral cricothyroid muscle and consequently the patient loses three characteristics of his voice. He gets easy fatigability of his voice, loses the ability to produce high pitched sounds (female singing registers) as well as the high intensity sounds. The internal laryngeal nerve is also a branch of the superior laryngeal nerve and is a pure sensory nerve. It supplies the interior of the larynx down to the level of the vocal folds.

Again, the recurrent (inferior) laryngeal nerve (RLN) is a branch of the vagus nerve. On the right side, it arises at the level of the root of the neck and winds around the right subclavian artery. On the left side, it arises in the superior mediastinum and then winds around the aortic arch and ascends to the neck. In the neck it lies in the *Behr's triangle* where it runs in the tracheo-esophageal groove. In the groove it runs deep to the branches of the ITA in most of the cases. Sometimes, it runs in between the branches and rarely superficial to the main artery and its branches. On the right side and in a less than 2% of the population the nerve arises in the neck and is thus termed *non-recurrent recurrent laryngeal nerve*. All the reported cases with a non-recurrent recurrent laryngeal nerve on the left side of the neck had a situs inversus scenario!!!

The RLN is a mixed nerve. It is motor to the intrinsic laryngeal muscles except the cricothyroid muscle. It is sensory to the interior of the larynx from the level of the vocal folds downwards including the vocal folds themselves.

Injury of the RLN leads to different patterns of paralysis of the adductors and abductors of the vocal folds. This in turn leads to an abnormally wide or the more dangerous abnormally narrow rima glottidis. The patient suffers from one or more of a variety of complaints. These include choking, aspiration, inability to cough, hoarseness of voice, dyspnea on rest and/or effort, aphonia and stridor. The causes and mechanisms of injury of the RLNs as well as their consequences are beyond the scope of this chapter.

Parathyroid Glands (PTGs)

The upper PTGs arise from the fourth branchial arch. The lower PTGs together with the thymus arise from the third branchial arch. Obviously, they have to migrate a distance longer than the upper ones, so they are more likely to be ectopic. Their migration together with the thymus makes the superior mediastinum one of the commonest locations for such an ectopic ITA. Most of the blood supply to the PTGs comes from the ITA

Micro-Anatomy

The thyroid gland has a thin fibrous capsule from which septae divide the gland into lobes and lobules. The lobule is composed of follicles, each of which consists of follicular cells arranged in a circular pattern around the thyroglobulin filled center. In between the follicles are distributed the parafollicular C-cells, lymphocytes, blood vessels and septae.

1. APPLIED PHYSIOLOGY

- The thyroid gland is an endocrine gland that secretes thyroxine hormone in two forms well known as T3 and T4 directly into the blood. It is under the feedback control of the pituitary gland through the thyroid stimulating hormone (TSH). The pituitary gland, in turn, is under the feedback control of the hypothalamus through the thyrotropin-releasing hormone (TRH).
- The thyroid gland concentrates inorganic iodine from the blood against its concentration, which is about 100 times more inside the thyroid cells than in the blood. Then, through a number of chemical reactions, the iodide salt is organified into organic iodine, then linked to tyrosine, forming mono-iodotyrosine (MIT) and diiodotyrosine (DIT), then coupled forming T2, T3, and T4. The T2 has no metabolic pathway, so it is degraded once more. The other two hormones are combined to globulins forming thyroglobulin or colloid material stored in the center of each follicle. These chemical reactions are blocked at different points by certain foods as cabbage, cauliflower and others, but more important, by anti-thyroid drugs (ATDs).
- When needed, the thyroid gland releases its hormones into the blood after breaking the thyroglobulin into its two components through a step called proteolysis. When the free hormones are released into the blood, nearly 99% of them become again attached to plasma proteins, leaving only 1% as free hormones (FT3 and FT4), exerting all the metabolic effects of the thyroid hormones.
- It is important to know that the proteolysis step can be blocked by excess of iodine intake as happens intentionally in the pre-operative preparation of thyrotoxic patients, and as a side-effect of prolonged intake of lithium, which is a constituent of mood stabilizing agents and anti-psychotic medications.
- Under normal physiological conditions, the three of these hormones TSH, T3 and T4 are all in normal ranges. In *hyperthyroid* states, T3 and T4 rise significantly in the blood and TSH becomes almost undetectable. In *hypothyroid* states, T3 and T4 declines in blood and so TSH increases significantly.
- Since thyroid hormones are bound to plasma proteins, their total estimation is *not* accurate to diagnose the actual thyroid status. So no one should measure T3 and T4 any longer. Instead, free portions of both hormones are better detected in the serum. These are known as FT3 and FT4.

INVESTIGATIONS

Laboratory (Thyroid Function Tests)

1. TSH
2. FT3
3. FT4
4. Anti-thyroglobulin antibodies
5. Anti-peroxidase antibodies
6. TSH receptor antibodies (Trab)

Imaging

1. Plain X-ray neck and chest.
2. Barium swallow
3. Ultrasound (US) neck (the most important).
4. Computed tomography (CT) scan of the neck.
5. Magnetic resonance imaging (MRI) of the neck.

Radioisotope studies

1. Thyroid scanning.
2. Whole body scanning (WBS)

Others

1. *Biopsy / FNAC*: Extremely in diagnosis if sonographically indicated.
2. *Laryngoscopy*: Important for assessment of the vocal cords

Common (Routine) Investigations

1. TSH, FT3 and FT4.
2. Thyroid auto-antibodies, if Hashimoto's thyroiditis is to be excluded.
3. US ± FNAC.
4. Others: tailored according to each case

SURGICAL TREATMENT OF THYROID DISEASES

Indications of Thyroidectomy

- Cosmesis
- Compression (dyspnea - dysphagia)
- Complications such as hemorrhage
- Retrosternal extension
- Cancer or suspicion of cancer

Types of Thyroidectomy (Extent ???)

- Total thyroidectomy
- Near total thyroidectomy
- Hemi-thyroidectomy
- Isthmectomy

Types of Thyroidectomy (When???)

- Elective thyroidectomy
- Prophylactic thyroidectomy
- Incidental thyroidectomy
- Emergency thyroidectomy

Complications of Thyroidectomy

Early Complications

- Stridor
- RLN damage
- Laryngeal edema
- Laryngeal spasm
- Tracheomalacia
- Hematoma
- Tracheal injury
- Esophageal injury
- Thyroid crisis
- Vomit aspiration
- Pain
- Dysphagia

Delayed Complications

- Hematoma
- Seroma
- Infection (cellulitis – infected collection)
- Hypocalcemia
- Ugly scar
- Voice changes
- Stridor

Post-operative Care

- Hospital stay: how long? The standard is 24 hours
- Hospital medications: mainly pain-killers (analgesics)
- Drain removal: when? The standard is 1-3 days
- Take home medications: The standard is analgesics, calcium and Vitamin D replacement in addition to scar medications for cosmesis.
- Post-operative instructions: Stitch removal: when? The standard is 5-7 days

Follow-up Regimen for Non-Neoplastic Goiters

- Must we give replacement therapy? No, if he did hemi-thyroidectomy. Yes, if he did total thyroidectomy.
- When to start replacement therapy? Following issuing of the pathology report.
- How much to replace? The standard is to start with 100 micrograms daily.
- How frequent to check hormonal blood levels? After 2 months, then annually.
- What parameters to check? TSH + FT4.

Follow-up Regimen for Neoplastic Goiters

- TSH + FT4 and:
- Thyroid tumor markers:
 1. Thyroglobulin
 2. Anti-thyroglobulin antibodies
 3. Calcitonin
 4. Carcino-embryonic antigen (CEA)
- Neck ultrasound
- Whole body scan (WBS).

CHAPTER

2

Clinical Approach to Goiter

Definition of Goiter

Goiter is an *enlarged thyroid gland irrespective of the cause or function*. Normally, the gland is 20-25 g in weight & is *neither visible nor palpable*. However, the *isthmus* is sometimes apparent, particularly during swallowing & can be felt in very thin persons.

Characteristic Clinical Features of a Thyroid Swelling

1. *Anatomical site*: Lower anterior part of the neck, *deep* to the sternomastoid muscle.
2. *Shape*: Butterfly; however, enlargement may be unilateral or asymmetrical.
3. *Mobility with deglutition*: A goiter moves up & down with deglutition.

Clinical Types of Goiter (Causes of Thyroid Enlargement) (Table 1)

Simple Goiter	Toxic Goiter	Special Goiter
1. Diffuse: a) Physiological b) Colloidal) 2. Multinodular 3. Solitary Nodule 4. Recurrent Nodular.	1. Diffuse (1 ^{ry}) = <i>Grave's Disease</i> 2. Multinodular (2 ^{ry}) = <i>Marine Lenhart Syndrome</i> . 3. Solitary Nodule = <i>Plummer's Disease</i> . 4. Recurrent Nodular.	1. Thyroiditis. 2. Neoplastic: a) Benign: Adenoma. b) Malignant: 1 ^{ry} or 2 ^{ry} . 3. Autoimmune. 4. Congenital (dysmorphogenesis).

N.B.

Nodules may be seen as *single lesions (uni-nodular)*, or several nodules may coalesce & remain segregated *in one lobe* of the thyroid gland (*pluri-nodular*), or several nodules of different sizes may be scattered irregularly *throughout the thyroid gland (multi-nodular)*.

HISTORY-TAKING

Personal History (Personal Data)

Age

- Young Age Physiologic goiter, papillary carcinoma.
- Between 25-40 y SNG - 1st thyrotoxicosis.
- Between 30-45 y 2nd thyrotoxicosis.
- Elderly Cancer of the thyroid.

Gender

- Generally, females are more commonly affected with goiter than males.
- Males are more affected with retro-sternal goiter.
- Males with a solitary thyroid nodule are more affected with thyroid cancer than females.

Symptoms (Complaints)

A. Symptoms due to thyroid swelling

1. Lump in the neck which is usually discovered accidentally & is slow-growing causing *disfigurement*. *Sudden appearance or rapid $\uparrow$ in size may be due to:* hemorrhage in a cyst or a necrotic nodule, rapidly growing carcinoma, or subacute thyroiditis.
2. Pain due to the same causes of rapid enlargement

B. Pressure symptoms: On 2 tubes (esophagus/trachea), 2 nerves (RLN/sympathetic trunk) & 2 vessels (carotid artery/IJV) :

1. *Dysphagia*: Because the thyroid has to be pulled upwards during swallowing.
2. *Dyspnea*: due to pressure over the trachea, especially if retro-sternal or malignant.
3. *Hoarseness of voice*, due to recurrent laryngeal nerve (RLN) affection in cancer.
4. *Horner's syndrome* due to affection of cervical sympathetic trunk in cancer thyroid.
5. Carotid artery causing dizziness & fainting attacks (rare).
6. IJV causing congestion of the face.

C. Endocrine symptoms

1. *Eye symptoms*: Staring look, difficulty in closing the eye due to proptosis, or even diplopia in advanced cases of thyrotoxicosis.
2. *Symptoms of toxicity*: Dyspnea on effort, tiredness, palpitation, intolerance to hot weather, $\downarrow$ weight despite $\downarrow$ appetite, diarrhea, nervousness & irritability, and menstrual disturbances.
3. *Symptoms of hypothyroidism (myxedema)*: $\downarrow$ in weight, Puffy eyelids with waxy yellow complexion, intolerance to cold weather, slow thought & speech, apathy, easy fatigability, loss of hair & constipation.

Present History

A) Onset

- The lump is usually accidentally discovered (ask about stress).
- Onset of toxic symptoms in relation to onset of lump: In 1st thyrotoxicosis, both appear simultaneously. In 2nd thyrotoxicosis, the lump appears first.
- Subacute thyroiditis has an acute onset.

B) Progress

- Slowly-progressing Simple and toxic goiters
- Rapidly-progressing Cancer thyroid, especially anaplastic type
- Self-limiting (within 1-3 m) Subacute thyroiditis.
- Sudden increase in size Suggests hemorrhage in a cyst, infection or cancer
- Ask about change in mobility Restricted (e.g. cancer).
- Change in consistency Harder (malignant transformation or calcification)

C) Course of treatment

You should determine the following:

- Type of drugs received by the patient (e.g. Lugol's I₂, Carbimazole, etc).
- The duration of treatment.
- Effect of treatment on the swelling or condition of the patient.

Past History

- *Similar condition*: SNG may turn 2nd toxic or malignant.
- *Intake of goitrogens* e.g. *drugs* (Thiocyanates for hypertension) or *food* (cabbage)
- *Previous irradiation* of the neck or the scalp may predispose to cancer thyroid.
- *Previous thyroid operation* (Recurrence or fixation / limited mobility).

Family History

- Family history of thyroid disease is important in endemic goiter, Hashimoto's disease (autoimmune), medullary carcinoma of the thyroid & dysmorphogenesis.
- Goiter + deafness in **Pendred's disease** = hereditary goiter.

Special Habits

- *Diet e.g. colloidal goiter*: Some vegetables contain chemicals which are *goitrogens* i.e. they interfere with hormone synthesis. An excess of cabbage in diet can cause goiter.

CLINICAL EXAMINATION

General Examination

General Condition

- Is the patient thin or fat ? Wasting ? Myxedema ?
- Evidence of hyperfunction ? Staring, excitable, trembling, nervous, sweating ?
- Evidence of hypofunction ? Slow thinking, dullness, apathy, puffy eyelids, etc ?

Vital Signs

- *Heart rate and rhythm:* Tachycardia (which persists during sleep - to exclude neurosis) suggests thyrotoxicosis (< 90/min = mild, 90-110/min = moderate, > 110/min = severe toxicity). Extra-systoles, atrial flutter or fibrillation may occur in severe thyrotoxicosis.
- *Temperature:* May be slightly elevated in thyrotoxicosis.
- *Blood pressure:* There may be systolic hypertension & ↑ pulse pressure in thyrotoxicosis
- *Respiratory rate:* Dyspnea at rest in toxic heart failure or in retro-sternal goiter.

Head and Neck

- *Skull:* For hard masses (metastases) in late cancer goiter.
- *Scalp and face:* Malar flush & scanty dry head hair (myxedema) - *Tongue*
 1. Tremors of protruded tongue (1^{ty} toxic goiter).
 2. Cyanosis (toxic heart failure).
 3. Ectopic lingual thyroid (a tongue lump – Figure 1).



Figure 1: Lingual thyroid

- *Neck*
 1. *Congested neck veins* in thyrotoxic heart failure (pulsating) and in retro-sternal goiter (non-pulsating).
 2. *Cervical lymphadenopathy:* hard in malignant goiter.
 3. *Position of trachea* (By feeling with the finger-tip in the supra-sternal notch).

The Eyes

- *Look for Horner's syndrome (ptosis, myosis, and enophthalmos)* in thyroid cancer with infiltration of the cervical sympathetic trunk (Figure 2).
- *Look for signs of toxicity*
 1. *Lid retraction (Dalrymple's sign)*
The upper eyelid is higher than normal, but there is no exophthalmos (Figure 3).
 2. *Stellwag's sign*
Staring look due to infrequent blinking (Figure 4)

3. *Lid lag (Von Graefe's sign)*

The patient is asked to follow the finger moved up & down. The upper lid does not keep pace with the moving eyeball (Figure 5).

4. *Lack of convergence (Moebius sign)*

The patient is asked to follow the finger while approaching the root of the nose. There is difficulty in convergence when the finger comes near to the eye.

5. *Joffroy's sign*

The patient can look upwards without wrinkling of the forehead).



Figure 2. Horner's syndrome of right eye - Note ptosis on right side



Figure 3. Upper lid retraction of left eye, with staring look.



Figure 4. Lid lag (the upper lid lags when the patient looks upwards)

- *Exophthalmos*

The sclera becomes visible below (Figure 5) or all around the iris (Figure 6).

Naffziger's Method: Stand behind the seated patient & tilt his head backwards, holding in a manner that keeps the hair out of the way. Observe the eyeballs; your plane of vision being that of the super-ciliary ridges. If no proptosis is noticed, then the staring look of the patient is due to lid retraction & not due to true exophthalmos.



Figure 5. Exophthalmos. Note visible sclera below the iris



Figure 6. Severe exophthalmos. Note visible sclera all around the iris

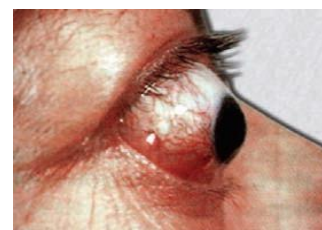


Figure 7. Exophthalmos + chemosis

- *Chemosis* (Figure 7), *Ophthalmoplegia* and dilated congested conjunctival blood vessels, in severe cases of thyrotoxicosis.

- Always note the relations of the iris to the sclera for proper differentiation between lid retraction & exophthalmos (Figure 8).

- *Tremors of the eyelids* in semi-closed eyes.

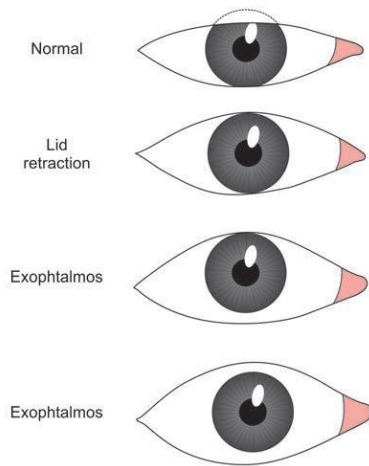


Figure 8. Relations of the iris to the sclera

The Heart

- Examine the heart for signs of heart failure (in 2^{ty} toxicosis).

The Chest

- Dullness over the manubrium (percussion) Retro-sternal goiter.
- Basal dullness rising towards axilla (percussion) Pleural effusion (lung metastases).
- Rhonchi and crepitations (auscultation) Lung metastases in cancer thyroid.
- Basal Crepitations (auscultation) Heart failure (2^{ty} thyrotoxicosis)

The Abdomen

- Hepatomegaly e.g. in late malignant goiter (hard and irregular).

Nervous System

- Bilateral exaggerated reflexes (ankles and knees) in thyrotoxicosis.

The Hands

- *Moist and warm* in thyrotoxicosis.
- *Fine tremors* in thyrotoxicosis: Ask the patient to hold her arms out in front of her, elbows and wrists straight, fingers straight and separated.

Lower Limbs

- *Muscle wasting* and toxic myopathy (in toxic goiter).
- *Edema* (heart failure) or *peri-tibial myxedema* (Figure 9), which is slightly pitting with orange chins at the beginning, but becomes non-pitting later on, with deep purple chins
- *Tender masses* in long bones (metastases in late cancer thyroid).



Figure 9. Bilateral myxedema

Local Examination

Position

- The patient is examined while sitting in a chair.
- The doctor stands in front of the patient during inspection, then behind during palpation.

Inspection

- Look at the neck for the presence of **swelling** (Figures 9-14) and note the number, site, size, extent, shape, color, skin overlying the swelling, pulsations and any pressure effects.



Figure 9: Graves' disease



Figure 10: Multinodular goiter (MNG)

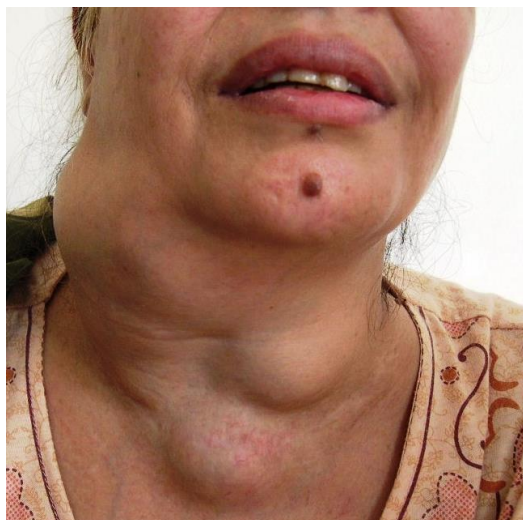


Figure 11: Huge multinodular goiter-MNG



Figure 12: Recurrent goiter. Note previous scar



Figure 13: Follicular carcinoma.
Note the dilated veins (arrow)



Figure 14: Anaplastic carcinoma

- In obese and bull-necked individuals, inspection of the thyroid is made easier, by the patient throwing her head backwards, and pressing her occiput against her clasped hands (*Pizzillo*).
- **Look for dilated veins** (thyrotoxicosis, malignancy and retro-sternal goiter), or **pulsations** (thyrotoxicosis), face **congestion** and cyanosis (retrosternal goiter).
- **Does the swelling move with deglutition?**
- Ask the patient to swallow. A thyroid swelling moves upwards with deglutition, but it may be limited by malignancy, inflammation, irradiation, scar of previous operation, or retrosternal extension.
- **Does it move upwards with protrusion of the tongue?**
- Ask the patient to open her mouth and protrude her tongue. A *thyroglossal cyst* does

Palpation

- *Check the position of the trachea* from the front
- *Comment on the Swelling:* Stand behind the patient, your thumbs on the scalp tilting the head forwards. Palpate by the other four fingers of each hand (Figure 15). A normal thyroid is not palpable. If palpable comment on:
 1. Is the whole thyroid enlarged?
 2. Is there is one nodule or more? Smooth enlargement of one lobe of the thyroid gland is considered as single nodule
- Comment on local temperature and tenderness; confirm information obtained by inspection (site, size, etc), surface, edge, consistency, thrill & mobility (horizontally and vertically).
- *Confirm mobility with deglutition.*
- *Try to get below the swelling.* Is there an extension into the mediastinum (retro-sternal)?



Figure 15: Palpation of goiter from behind

- *Pulsations of both carotids* are felt and compared (Figure 16).
- *Cervical lymph nodes (LNs)* are palpated.



Figure 16:
Seeking
pulsation of
carotid artery

- *Berry's Sign*

When the gland enlarges, it displaces the carotid tree backwards and outwards, therefore in many cases the pulsation of the carotid artery can be felt behind the posterior edge of the swelling. In malignancy, the artery tends to become surrounded by the tumor.

Percussion

Percuss over the sternum for retro-sternal extension (dull). This is confirmed by plain X-ray & CT-scan. The thyroid nature of the dullness is confirmed by thyroid scanning.

Auscultation

Auscultation over a thyrotoxic goiter may reveal a bruit.

INVESTIGATIONS

- Routine laboratory tests.
- *Thyroid Function Tests* (T3, T4, TSH), are done in doubtful cases and interpreted in light of clinical findings.
- Thyroid antibodies (anti-TPO, anti-Tg) & serum calcitonin in selected case
- *Plain X-Ray* of the neck & upper chest: Soft tissue shadow (retrosternal goiter-RSG) (Figure 17).
- *CT scan*: for RSG (Figure 18), anatomical relations and cervical LNs.
- *US* can diagnose cystic from solid nodules, detect clinically impalpable nodules, identify suspicious nodules and guide fine needle aspiration cytology (FNAC)-for more accurate diagnosis.
- *Thyroid scanning* is useful in case of a solitary thyroid nodule (STN) (It can show a cold or hot nodule).
- *Indirect Laryngoscopy* (for assessment of the vocal cords).
- *Biopsy (FNAC or Tru-cut needle biopsy)*: in case of suspected malignancy.



Figure 17: PXR chest showing retrosternal shadow **Figure 18.** CT scan showing retrosternal shadow

DIAGNOSIS & DIFFERENTIAL DIAGNOSIS

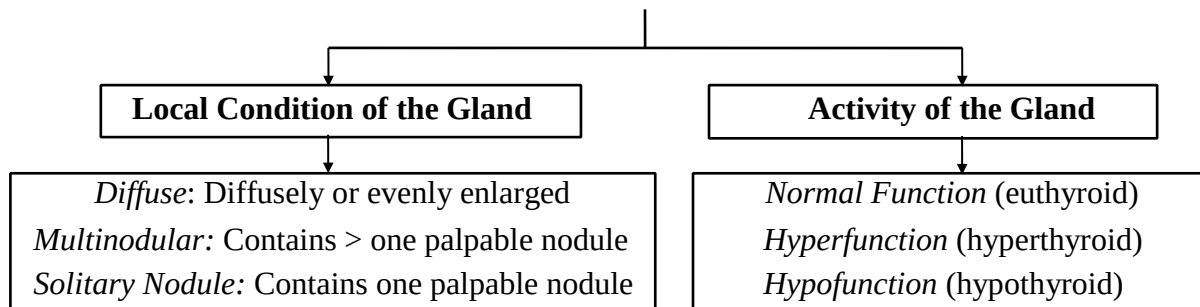
Is it Goiter or not ?

A thyroid swelling has the following characteristics:

1. *Anatomical Site*: Lower anterior part of the neck, *deep* to the sternomastoid muscle.
2. *Shape*: Butterfly (2 lobes + isthmus). However, it may be unilateral or asymmetrical.
3. *Mobility with deglutition*: A goiter moves up & down with deglutition. However, mobility may be restricted in certain cases (refer back). Also, remember that *not every swelling mobile with deglutition is a goiter*. Some other swellings do.

Which Type of Goiter ? Simple - Toxic - Malignant ??

Demonstrating after examination that the swelling is an enlarged thyroid, you should be able to conclude the following information about the gland



- *Type of goiter:* This is reached by determining the state of activity of the gland, and its local condition (diffuse, nodular or one nodule).
- *The state of activity of the gland* is determined, whether normal (simple goiter), hyperthyroid (toxic goiter; 1^{ry} or 2^{ry}), or hypothyroid (myxedema). This is usually reached through history & general examination.
- *Is goiter malignant?* Malignancy could be suspected from the rapid rate of growth, presence of pain, hoarseness of voice, pressure symptoms, absent or weak carotid pulsation and enlarged cervical LNs.
- Is the swelling obstructing the trachea causing stridor? (rare but vital).
- Does the swelling extend behind the sternum?

Is There Retro-sternal Extension?

- *History:* Postural dyspnea, stridor, cough, wheezing, choking, dysphagia, etc.
- *Physical examination* (Figure 19)
 1. Dilated veins in front of the neck and sternum.
 2. Enlarged thyroid with non-visible lower border on swallowing (inspection).
 3. Impalpable lower border on palpation.
 4. Dullness on percussion over the manubrium sterni.



Figure 19: Huge retrosternal goiter (RSG)

5. Flushing of the skin & dilatation of the EJV during raising the arms or hyperextension of the neck (= *Positive Pemberton' Sign*).
6. *Investigations*
 - a) Plain X-ray.
 - b) CT scan.
 - c) Thyroid scan.

Final Diagnosis (Tables 2-4)

After determining whether it is simple, toxic or malignant, the local condition of the gland is assessed (Table 2):

- Is it diffusely enlarged (physiologic, colloid, 1^{ry} thyrotoxicosis, or thyroiditis)?
- Is it diffusely nodular (Simple nodular, 2^{ry} thyrotoxicosis)?
- Is it one nodule which is felt (Solitary adenoma, cyst, toxic nodule, or malignant nodule)?
- Malignant goiter is hard, may be fixed, and there may be cervical LN metastases.

Table 2: How to reach final diagnosis

Local condition of the gland	Toxic symptoms - Present or Not ?	
	Simple	Toxic
- Diffusely enlarged - Diffusely nodular: - One nodule (solitary)	Physiologic, colloid, Thyroiditis SNG, malignant	1 ^{ry} Thyrotoxicosis 2 ^{ry} Thyrotoxicosis Toxic nodule.

Table 3: Differences between 1^{ry} and 2^{ry} thyrotoxicosis

Point of Difference	1 ^{ry} Thyrotoxicosis	2 ^{ry} Thyrotoxicosis
- Age	Young adults	Older age group
- Onset of toxic symptoms	Simultaneous with swelling	Follows the swelling
- Eye signs	+++	+
- Nervous manifestations	++++	+
- Cardiovascular manifestations	+++	++++
- GIT manifestations	+++	+
- Increased BMR manifestations	+++	+
- The thyroid gland	Diffuse and smooth, symmetrical, bilateral, fleshy	Nodular, asymmetrical, may be unilateral, firm

Table 4: Differences between Grave's disease, colloid goiter and thyroiditis

Criteria	1 ^{ry} Toxic (Grave's)	Colloid Goiter	Thyroiditis
- Size	Slight to moderate	Moderate to gross	Small or moderate
- Surface	Smooth	Bosselated	Smooth
- Consistency	Soft-Fleshy	Fleshy	Hard
- Tenderness	-	-	+
- Bruit	+	-	-

CHAPTER

3

Thyroid Tumors

BENIGN TUMORS OF THE THYROID

Pathological Classification (Table 1 below)

Epithelial Tumors	Mesenchymal Tumors	Other Tumors
1. Papillary adenoma (fetal or microfollicular adenoma). 2. Follicular adenoma (cystadenoma or colloid adenoma).	1. Lipoma 2. Leiomyoma 3. Hemangioma.	Teratoma (mainly in children).

Characteristic Features of Follicular and Papillary Adenomas (Table 2 below)

Features	Follicular Adenoma (Macrofollicular)	Papillary Adenoma (Microfollicular)
<i>Incidence</i>	More common	Rare
<i>Age</i>	Older age	Younger (18-20 years - not since birth)
<i>Size</i>	Any size	Usually remains small
<i>Site</i>	Any part of the gland	Usually at junction of isthmus with one lobe
<i>Function</i>	Euthyroid, but may be <i>toxic adenoma</i> .	Euthyroid
<i>NEA</i>	Apparently encapsulated but the surrounding tissue is usually the seat of tiny adenomas a. Cut section may be solid or cystic (<i>cystadenoma</i>).	It is probably a true tumor and the rest of the gland is normal. Cut section is solid + formed capsule from which radiating septa pass to the nodule
<i>MP</i>	Large acini full of colloid with islets of hyperplasia + hemorrhage, necrosis, fibrosis, calcification, ossification, etc).	Cuboidal cells arranged in strands with no or little acinar structures. It is called "fetal" because it simulates the fetal thyroid tissue.
<i>Differential Diagnosis</i>	Nodular hyperplasia – Minimally-invasive FTC - Follicular pattern of PTC	Other types of adenomata.
<i>Treatment</i>	Hemi-thyroidectomy because resection-enucleation has a high recurrence rate.	Hemi-thyroidectomy is always preferable.

FTC: follicular thyroid carcinoma, PTC: papillary thyroid carcinoma

MALIGNANT TUMORS OF THE THYROID

When to be Suspected?

Thyroid cancer is suspected in the presence of *an enlarging painless lesion* with one or more of the following:

- Radiation exposure
- Male gender, older age, younger age
- Rapid ↑ in size
- Previous thyroid cancer
- Lymphadenopathy
- Evidence of local invasion (vocal cord paralysis, dysphagia or firm, fixed nodules)
- Familial syndromes (Table 3)

Table 3: Familial syndromes: Familial non-medullary thyroid cancer

Syndrome	Thyroid cancer	Description
<i>Gardner syndrome</i>	PTC	Intestinal polyps, osteomas, fibromas, lipomas
<i>Cowden syndrome</i>	PTC - FTC	Breast cancer, hamartomas
<i>Carney syndrome</i>	PTC	Myoma, Schwannoma, pigmented adrenal nodules, pituitary adenomas, testicular tumors

PTC: papillary thyroid carcinoma – FTC: Follicular thyroid carcinoma

Incidence

- | | |
|---|----------------------|
| - Papillary or papillary/follicular = 80% | - Anaplastic = 5-15% |
| - Follicular = 15% | - Lymphoma = rare |
| - Medullary = 2-10% | - Metastatic = rare |

Classification

- *Differentiated tumors of follicular origin (90-95 %):* Papillary carcinoma, follicular carcinoma, Hurtle cell carcinoma
- *From parafollicular cells (2-10%):* Medullary thyroid carcinoma (MTC)
- *Poorly differentiated (5-15%):* Anaplastic thyroid carcinoma (ATC)

Oncogenes associated with Thyroid Carcinoma

1. RET oncogene: Papillary (PTC) & MTC.
2. Mutated RAS oncogene: Follicular thyroid carcinoma (FTC)
3. Mutated p53 gene: Anaplastic thyroid carcinoma (ATC).

PAPILLARY THYROID CARCINOMA(PTC)

Incidence

- It is the most common histological variety of thyroid malignancy (80%) and is considered the predominant thyroid cancer in children.
- May be due to radiation exposure of the neck.
- *Age*: Peak incidence is in the third decade of life.
- *Gender* (female : male = 3:1)

Pathology

- It is composed of complex papillary projections with a fibrovascular core (Figure 1). *The hall mark diagnostic features are Psammoma bodies* (laminated calcified spheres) (Figure 2) and *Orphan Annie eye Nuclei*_or ground glass nuclei (nuclei that contain finely dispersed chromatin, which imparts an *optically clear or empty* appearance) (Figures 3 and 4).
- The incidence of multi-focality is 80%
- It has a propensity to spread to LNs in 30-50% of patients, albeit with no effect on survival. Hematogenous spread is late to lung & bones.

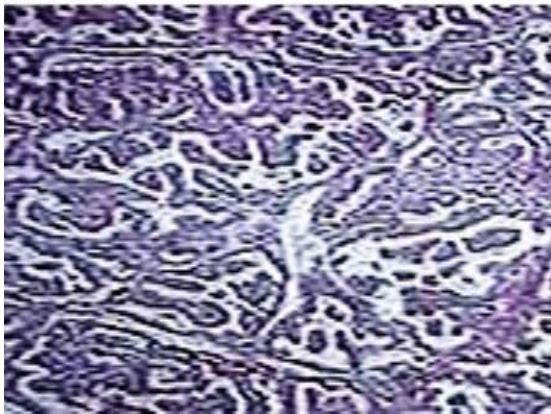


Figure 1: PTC (low resolution). Notice the frond-like projections

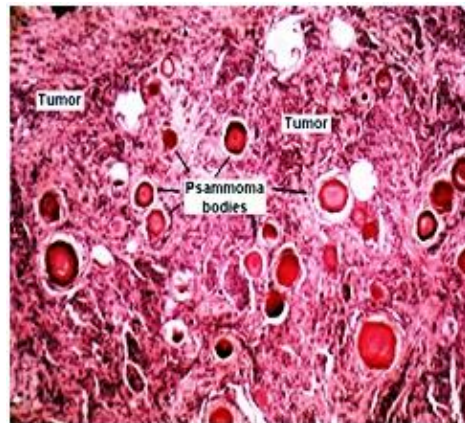


Figure 2: PTC: Psammoma Bodies (Laminated Calcific Densities)

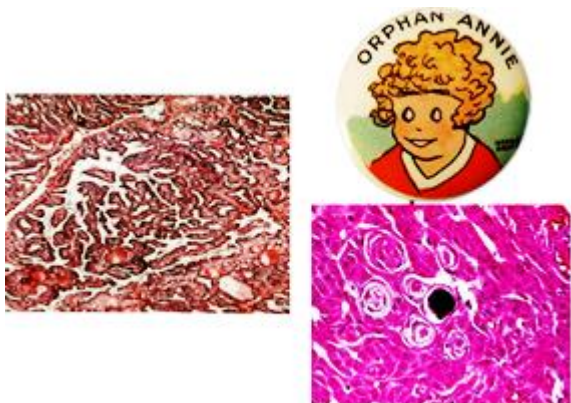


Figure 3: PTC: Papillary formation and orphan Annie sign

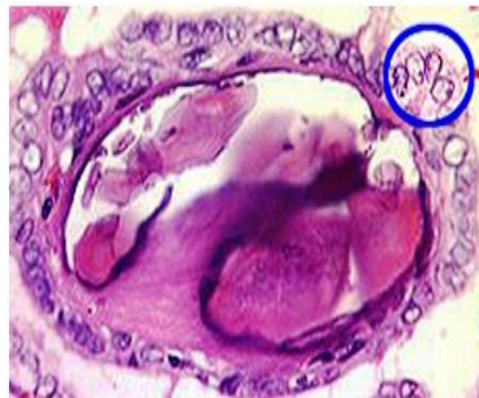


Figure 4: Psammoma body within a PTC. Large clear areas within the nuclei looking like "Little Orphan Annie" eyes (circled in blue)

- *Clinico-pathological Forms of PTC*

Based on tumor size and extent, 3 clinico-pathological forms of PTC are recognized, all of which *may be associated with LN metastases and intra-thyroidal blood vessel invasion or occasionally metastases.*

1. *Minimal or occult / micro carcinoma*: 1 cm or less, no capsular invasion, on-palpable and usually an incidental finding or at autopsy.
2. *Intra-thyroidal tumors*: >1cm, confined to the thyroid gland with no extra-thyroidal invasion
3. *Extra-thyroidal tumors*: Locally advanced with invasion through the capsule into adjacent tissues

Diagnosis

Symptoms and signs

- Euthyroid, slowly growing painless mass
- Manifestations of local invasions (late): Dysphagia, dyspnea, hoarseness of voice
- Palpable cervical LN: More apparent than primary lesion (*occult tumor*)
- Distant metastases (uncommon): lung metastases in *children*

Investigations

- **Ultrasonography (US)**

Sonographic features that are helpful are:

1. *Calcifications*: Thyroid microcalcifications, which are Psammoma bodies are 10–100-µm round laminar crystalline calcific deposits. They are one of the most specific features of thyroid malignancy, with a specificity of 85%.
2. *Local invasion* of adjacent soft tissue & *LN metastases* are highly specific of thyroid cancer. The US features that raise suspicion of LN metastases include a rounded bulging shape, ↑ size, replaced fatty hilum, irregular margins, heterogeneous echo-texture, calcifications, cystic areas & vascularity throughout the LN instead of normal central hilar vessels at Doppler imaging
3. *Shape*: The shape of a thyroid nodule is a potentially useful US feature in that a solid thyroid nodule that is taller than it is wide (i.e. greater in its antero-posterior dimension than its transverse dimension) has a 90% specificity for malignancy.
4. *Vascularity* (color or power Doppler US): The most common pattern of vascularity in thyroid malignancy is marked intrinsic hyper-vascularity, which is defined as flow in the central part of the tumor that is greater than that in the surrounding thyroid parenchyma
5. *Hypoechoic solid nodule*: Malignant nodules, both carcinoma & lymphoma, typically appear solid & hypoechoic when compared with normal thyroid parenchyma.

- **FNAC** (specific and sensitive for PTC, MTC and ATC)
- **CT/MRI** in patients with extensive local or sub-sternal extension

Surgery

- Hemithyroidectomy (lobectomy with isthmectomy) is acceptable for minimal PTC.
- Total thyroidectomy if (1) size >4cm, (2) age (male >40 y, female >50y) & angio-invasion
- Total thyroidectomy + Neck dissection if there is a proof of metastatic cervical LNs

Reasons for total thyroidectomy:

1. 80% is multifocal
2. To ↓ incidence of anaplasia in any residual tissue
3. Facilitate the Dx of unsuspected metastatic disease by RAI scanning or treatment.
4. Greater sensitivity of blood thyroglobulin level to predict recurrent or persistent disease.

FOLLICULAR THYROID CARCINOMA(FTC)

Incidence

- It is the 2nd common thyroid cancer (10%) and is more frequent in I₂-deficiency areas suggesting that nodular goiter may predispose to the development of the neoplasm
- High frequency of *RAS* mutations in follicular adenomas and carcinomas suggests that the two may be related tumors
- Age: Mean age is 50 y
- Gender: (female : male = 3:1)

Pathology

- Most FTCs are composed of uniform cells forming small follicles containing colloid appearance of normal thyroid. The following features should be taken into consideration:
 - Occasional tumors are dominated by cells with abundant granular, eosinophilic cytoplasm (*Hürthle* cells).
 - Psammoma bodies are *not* present
 - Nuclei lack the features typical of PTC.
 - Follicular lesions with nuclear features typical of PTC should be treated as PTC
 - Nuclear features are *not* helpful in distinguishing follicular adenomas from minimally invasive follicular carcinomas
 - Requires extensive sampling of the tumor-thyroid *capsule* interface to exclude capsular (Figure 5) or vascular (Figure 6) invasion.
 - Unlike in PTCs, lymphatic spread is distinctly uncommon in follicular cancers

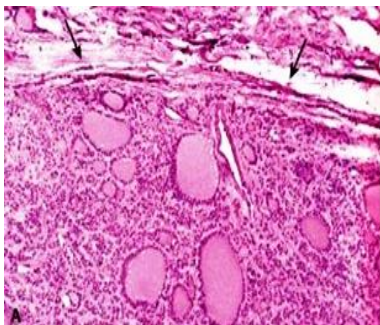


Figure 5: A. Follicular adenoma: Compressed thyroid tissue with no capsular invasion. **B.** FTC with capsular invasion (arrow)

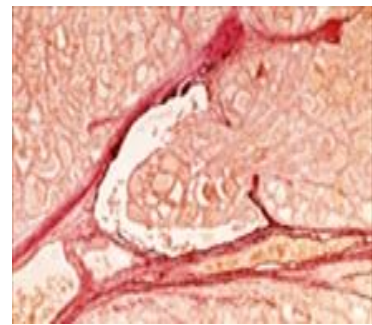
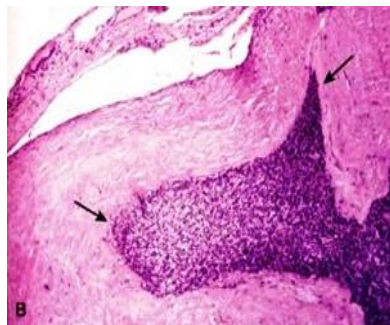


Figure 6: FTC with vascular invasion

- Thyrotoxic in 1% of cases
- It has a propensity to spread via the *blood* stream. Vascular invasion and hematogenous spread is more common (bone, lung and liver).
- Types:
 - A. Minimally-invasive tumor: Invasion into, but *not* through, the tumor capsule
 - B. Invasive tumors (capsular/vascular).

Diagnosis and Prognosis

- FNAC not helpful → lobectomy and isthmectomy , *Mortality*: 40 % (at 10 years)

Scoring and Prognosis of DTC (Differentiated Thyroid Cancer)

- AGES and AMES scoring systems: **A** (Age), **G** (Grade of tumor), **M** (Distant metastases), **E** (Extent of tumor), **S** (Size of tumor)
- Both scoring systems have identified 2 distinct subgroups;
- *Low-risk group*
 - o Men 40 y or younger, women 50 y or younger, without distant metastasis (bone and lungs)
 - o Older patients with intra-thyroid follicular / papillary carcinoma, with minor capsular involvement with tumors < 4cm in diameter
- *High –risk group*
 - o All patients with distant metastasis and older patients with extrathyroid papillary/follicular carcinoma and tumors >4 cm regardless of extent of disease

HURTHLE CELL THYROID TUMOR

- *Incidence:* 3-5%
- *Gender:* Male : Female (2:1)
- *Origin:* Derived from oxyphilic cells of the thyroid gland.
- *Possesses* TSH receptors
- *Spreads* by lymphatics Only 10% takes up iodine
- *Diagnosis:* FNAC (20% malignant) - Often multifocal and bilateral.
- *Treatment:* Total thyroidectomy + Modified radical neck dissection (if with palpable cervical LNs). Thyroid suppression is suggested
- *Prognosis:* 5-y-SR (survival rate): 60%

MEDULLARY THYROID CARCINOMA (MTC)

- *Incidence:* 5-7%
- *Age:* The peak incidence is at 50-60 years
- *Origin:* Arises from *parafollicular* or C cells of the thyroid (neuroectodermal- ultimobranchial bodies, 4th and 5th branchial pouches).
- Secretes calcitonin (95%); 85% secretes carcinoembryonic antigen (CEA)
- *Sporadic* 90%: Uni-focal, usually at 45y, worse prognosis.
- *Familial* 10%: Multifocal, usually 35 y, better prognosis, associated with:
 1. *MEN IIA or Sipples' syndrome* (MTC, hyperplastic parathyroid and pheochromocytoma)
 2. *MEN IIB* (MTC, pheochromocytoma, ganglioneuromatosis and Marfan's syndrome)
- Does not concentrate ¹³¹I
- Can secrete: *Calcitonin* (95%), histamine, serotonin (causes diarrhea), ACTH (2-4% causing Cushing syndrome), CEA (85%), Prostaglandin E2 and F2 alpha.
- *Spread:*
 1. Lymphatics (neck and superior mediastinum)
 2. Blood → liver, bone (osteoplastic) and lung
 3. Local invasion

- **Diagnosis:** Serum calcitonin, CEA, FNAC, histology shows a characteristic amyloid stroma (Figure 7).
- **Treatment**
 - Total thyroidectomy
 - MRND is done for palpable cervical LN, tumor >2cm as there is a chance of 60% nodal metastasis
 - Tumor debulking in cases of metastatic & local recurrence is done to ameliorate symptoms of flushing & diarrhea & help to ↓ the risk of death.
 - Radiotherapy and chemotherapy → failure
 - All patients should be screened for pheochromocytoma (MEN II) which should be resected first.
 - Selective removal of the parathyroid glands should be done if there is preoperative hypercalcemia.
- **Follow up:** Serum Calcitonin / CEA level
- **Prognosis:**
 - Localized → 10-y-SR of 80%, (+) LN → 10-y-SR of 45%
 - Best → worst prognosis: Familial non-MEN MTC → MEN IIA → sporadic cases → MEN IIB

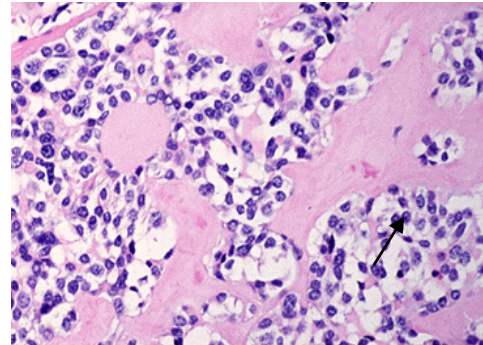


Figure 7. MTC: Amyloid stroma (arrow)

ANAPLASTIC THYROID CARCINOMA(ATC)

- **Incidence:** An uncommon thyroid malignancy affecting older patients
- **Origin:** May arise in a well differentiated thyroid carcinoma
- 80% of patients have a history of a long-standing goiter with sudden and rapid growth, hoarseness, dysphagia & airway compromise (Figure 8)
- Tracheal invasion (Figure 9) and/or bilateral recurrent nerve paralysis can be seen
- **Treatment**
 - Surgical excision is rarely possible without sacrificing essential cervical structures, but tissue diagnosis is needed to differentiate from lymphoma. Tracheostomy and total thyroidectomy are extremely difficult
 - External radiation may temporarily control the local effects of ATC.
- Limited effect of systemic chemotherapy (Adriamycin); no hormonal therapy known
- **Prognosis** is extremely poor with a mean life expectancy of 6-9 months. *Death* occurs from local invasion of vital cervical structures and airway compression



Figure 8: ATC with an inflammatory component



Figure 9: ATC infiltrating the trachea (white arrow)

THYROID LYMPHOMA

- Non-Hodgkin B-cell lymphoma
- Hashimoto's thyroiditis is a risk factor
- Compression manifestations
- *Treatment*
 1. Chemotherapy / Radiotherapy
 2. *Surgery*: It is curative when the disease is limited to the thyroid gland only
- *Prognosis*: 80% survival if confined to the gland; 40% if it had spread elsewhere

METASTATIC CARCINOMA

- Rare
- Hypernephroma is the most common primary site

CHAPTER

4

Neck Mass

Classification

- Neck masses can originate from: skin, endocrine organs, upper aerodigestive tract, vessels, or lymph nodes.
- They are classified into:
 1. Congenital
 2. Acquired
 - a) Inflammatory
 - b) Benign Neoplasm
 - c) Malignant Neoplasm

Evaluation

Evaluation, which leads to the proper treatment and the best outcome follows the following four steps:

1. Appropriate initial assessment
2. Role and technique of fine-needle aspiration biopsy (FNAB)
3. Appropriate use and interpretation of imaging
4. Management: Importance of specialized multidisciplinary care if malignancy is suspected

APPROPRIATE INITIAL ASSESSMENT

The correct diagnosis of a lump in the neck can often be made with a careful history and examination. The clinical signs of size, site, shape, consistency, fixation to skin or deep structures, pulsation, compressibility, trans-illumination or the presence of a bruit still remain as important as ever.

Age

- Children (Pediatric): Inflammatory – congenital - malignant
- Young adult: Congenital - inflammatory - malignant
- Adult >40 years: Malignant - congenital - inflammatory

Location

Common neck masses by 'location' include the following:

- Angle of mandible: parotid swelling
- Lateral neck: Enlarged lymph nodes (LNs)
- Central compartment: Thyroid swelling

Rule of '80' of Neck Masses

Causes of neck masses follow the rule of '80' as shown in Figure 1. As may be seen, 80% are neoplastic, of which 80% are malignant, of which 80% are metastatic.

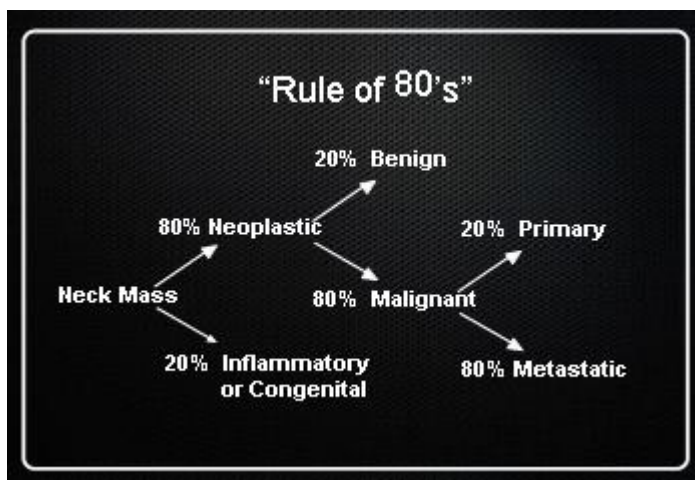


Figure 1: The rule of '80' of neck masses

Risk Factors

- *Sexual behavior*
- Male predominance: Cancer
- Younger patients
- Fewer traditional risk factors
- Sexual behavior as risk factor multiple sexual partners (>6) higher rates of oro-genital contact with multiple partners
- *Sun exposure*

Symptoms

- *Symptoms of Head and Neck Primary*
 1. Unilateral otalgia
 2. Unilateral hearing loss
 3. Nasal obstruction (snoring)
 4. Dysphagia
 5. Hoarseness
 6. Hemoptysis
 7. Epistaxis

- *Symptoms of Lymphoma*

1. Fever
2. Night Sweats
3. Weight Loss

Physical Examination

What do we need to document (Figures 2-8) ?

- Location of the mass in the neck
- Presence/absence of a primary in the head and neck
- Presence/absence of generalized lymphadenopathy

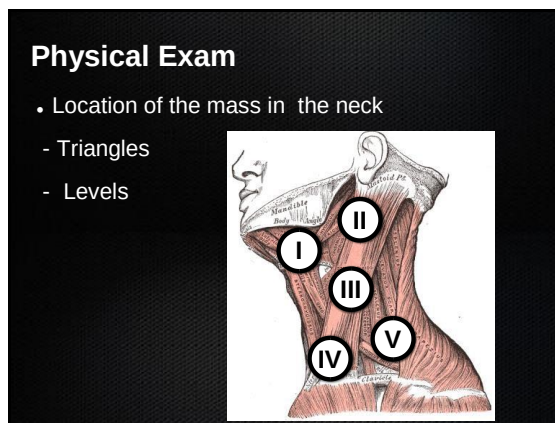


Figure 2: Triangles and levels of the neck

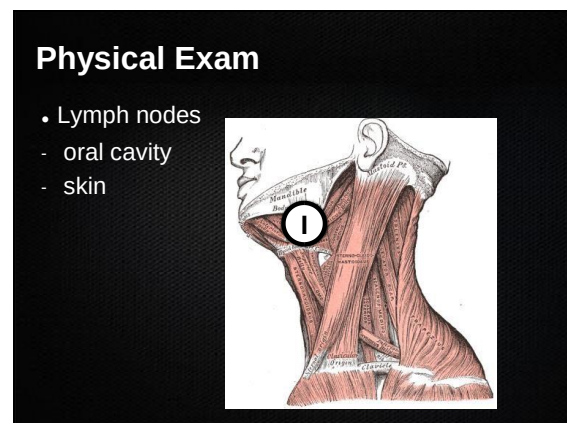


Figure 3: Level I neck masses

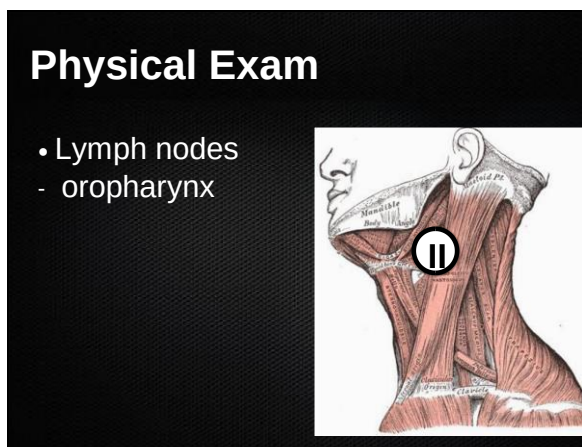


Figure 4: Level II neck masses

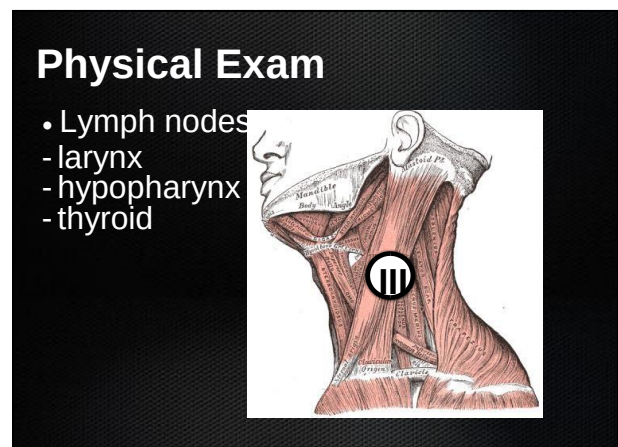
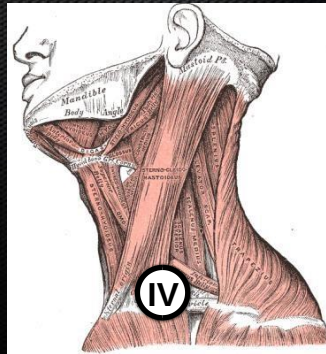


Figure 5: Level III neck masses

Physical Exam

- Lymph nodes
- Thyroid
- Below Clavicle



Physical Exam

- Lymph nodes
- nasopharynx

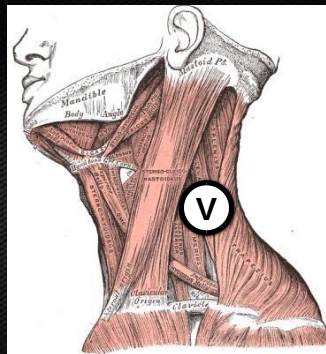
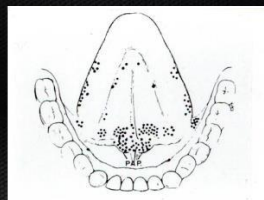


Figure 6: Neck masses at levels IV (above) and V (below)

Physical Exam

- Presence/absence of a primary in the head and neck- oralcavityand oropharynx

Distribution of
Early Oral Cancer



Mashberg. Cancer1973;32:1436-1445

Figure 7: Primary tumor in the neck

Physical Exam

- Presence/absence of a primary in the head and neck- oropharynx and larynx

Palpation Base of Tongue
FiberopticNasendoscopy

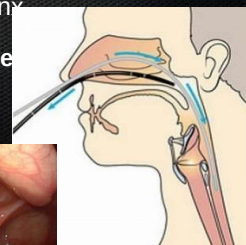


Figure 8: Primary tumor in pharynx and larynx

ROLE AND TECHNIQUE OF 'FNAB' / OPEN BIOPSY

FNAB

- Needle size: 25 gauge
- 12-15 Passes should be performed
- Immediate assessment of adequacy by the pathologist is the rule (Figure 9).

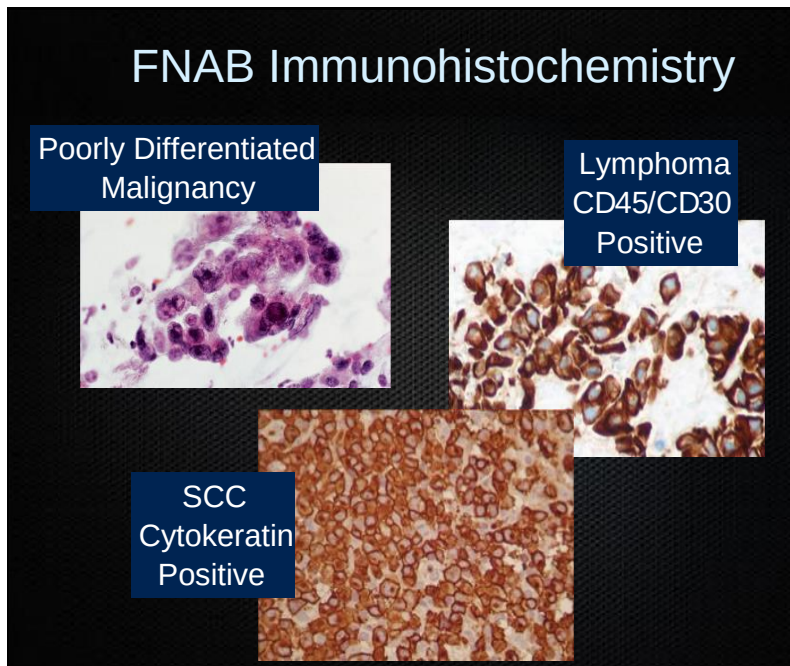


Figure 9: FNAB results of poorly differentiated carcinoma, lymphoma and squamous cell carcinoma (SCC)

Role of Open Lymph Node Biopsy

Excisional/Incisional biopsy may be necessary for:

- Sub-classification of lymphoma
- Persistently non-diagnostic FNAB
- Facilitating diagnosis of poorly differentiated carcinoma

3. IMAGING: APPROPRIATE USE AND INTERPRETATION

Appropriate use and interpretation are mandatory for reaching diagnosis of a neck mass as shown in Figure 10 and Table 1 below. The commonest imaging modalities used are ultrasonography (US), computed tomography (CT), magnetic resonance imaging (MRI) and positron emission tomography (PET)

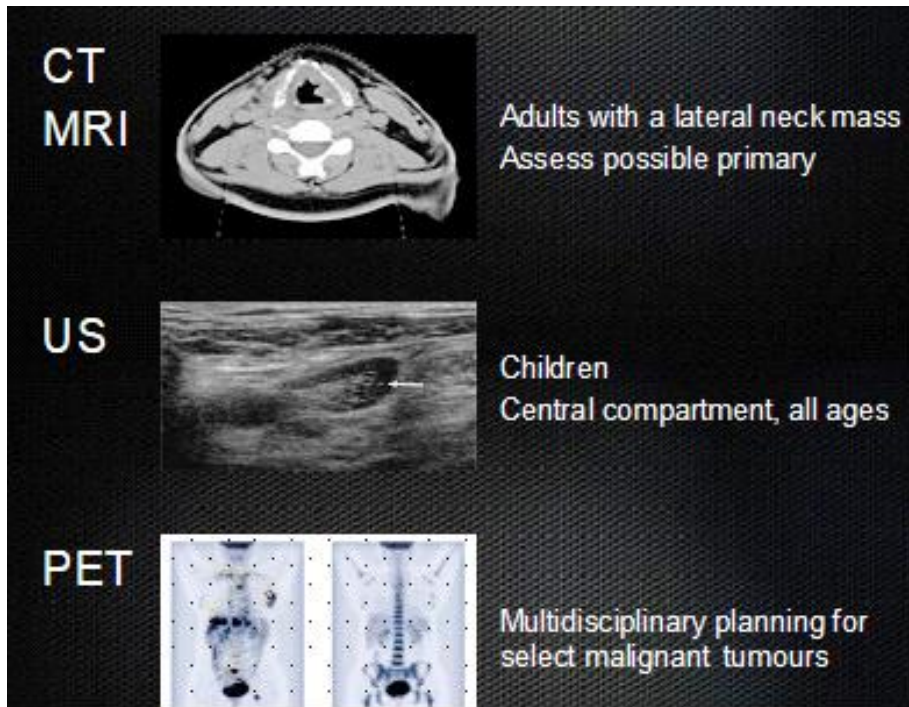


Figure 10:
Assessment of
neck masses with
different imaging
modalities.

Table 1: Recommended imaging modalities for neck masses in adults

Preferred Imaging Modalities for Neck Masses or Adenopathy in Adults		
Scenario	Recommended test	Comments
Nonpulsatile solitary mass or multiple neck masses	Contrast-enhanced CT	—
Pulsatile neck mass	Contrast-enhanced CT and contrast-enhanced CT angiography	May be performed at the same time
Patient with a neck mass and history of cancer treatment	Contrast-enhanced CT or CT with positron emission tomography	Positron emission tomography is superior in this subset of patients

ADEQUATE MANAGEMENT

Importance of specialized multidisciplinary care cannot be more emphasized, particularly if *malignancy* is suspected

NON-MALIGNANT NECK LUMPS

Cystic Hygroma (Lymphangioma)

- It is a congenital lesion, usually present within the first year of life.
- It occurs in the posterior triangle (Figure 11)
- Usually remains unchanged into adulthood
- Soft, cystic, multilocular, partially compressible and brilliantly trans-illuminant and may present with pressure effects
- CT or MRI may help define the extent of the neoplasm
- Treatment of lymphangiomas includes injection with picibanil or excision for easily accessible lesions or those affecting vital functions



Figure 11. Cystic hygroma

Branchial Cleft Cyst

- Remnant of branchial cleft (2nd)
- Most commonly occur in the second or third decades
- Pain +/- (severe throbbing pain)
- Usually presents as a smooth, fluctuant non-tender (tender), non-trans-illuminant mass mobile forwards and downwards, underlying the anterior border of the SCM muscle.
- Branchial fistula or sinus
- Primary treatment is with control of infection by antibiotics, followed by surgical excision.

Thyroglossal Cyst

- This is a common congenital midline neck mass (Figure 12 up)
- Sometimes at the lateral edge
- Pain and tenderness +/-
- Can be moved transversally
- Elevates on protrusion of the tongue.
- Treatment is with initial control of infection (if present) with antibiotics, followed by surgical excision including the mid-portion of the body of the hyoid bone (*Sistrunk's procedure*).
- Occasionally, these lesions become infected and resolve, or persist following drainage as a thyroglossal fistula (Figure 12 below)



Figure 12: Thyroglossal cyst and fistula

Lipoma

- Lipomas are the most common benign soft tissue neoplasms in the neck.
- They are poorly defined, soft masses, usually occurring after the 4th decade of life.
- They are usually asymptomatic, soft.
- FNAC or MRI scan can confirm the diagnosis.
- Surgery is indicated when the lump is increasing in size, for cosmesis, or when there is doubt about the accuracy of diagnosis.

Sebaceous Cyst

- These are common masses occurring often in older people but can occur at any age.
- They are slow growing, but sometimes fluctuant and painful when infected.
- Diagnosis is made clinically; the skin overlying the mass is adherent and a punctum is often identified.
- Excisional biopsy confirms the diagnosis.

Cervical Lymphadenopathy

Acute Lymphadenitis

- Tender swelling
- Antibiotic trial, less acute inflammatory nodes generally regress in size over 2–6 weeks.
- If the lesion does not respond, biopsy is warranted

Tuberculous (TB) Cervical Lymphadenitis

- Upper and middle deep cervical lymph nodes (LNs)
- Onset: gradually
- Pain (+/-). Systemic symptoms are unusual in young patients
- Abscess (painful, increase size, and skin discoloration)
- Mass: indistinct, firm, matted, fluctuate!
- Temperature is normal (*cold abscess*) (Figure 13)
- Treatment with anti-TB drugs for 6-9 months. Rifampicin, Ethambutol, INH, Pyrazinamide

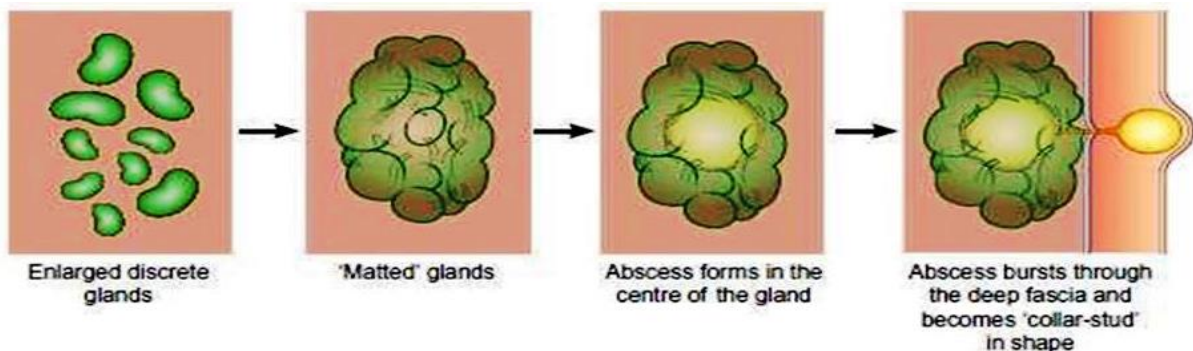


Figure 13: Consequences of tuberculous lymphadenitis

Carotid Body Tumor

- Rare tumor of chemo-receptors
- Age: 40-60 years.
- Slow-growing, painless, sometimes pulsating lump that may be bilateral (Figure 13).
- Side-to-side movement
- Symptoms of transient cerebral ischemia!
- Also known as *potato tumor* (hard, non-tender)
- Palpation may induce a vasovagal attack
- Biopsy is contraindicated.
- MRI may be utilized
- Angiography is the investigation of choice (Figure 13).
- Surgical removal is based on patient factors and symptoms.



Figure 13: Carotid body tumor

Pharyngeal Pouch

- Diverticulum of the pharynx through the gap between horizontal fibers of the crico-pharyngeus muscle below and the lowermost oblique fibers of the inferior constrictor muscle above.
- History of froth and acid taste
- Halitosis and regurgitation of food.
- There is no bile or to it.
- Pressure on the swelling causes gurgling sounds and regurgitation
- Treatment: cricopharyngeal myotomy

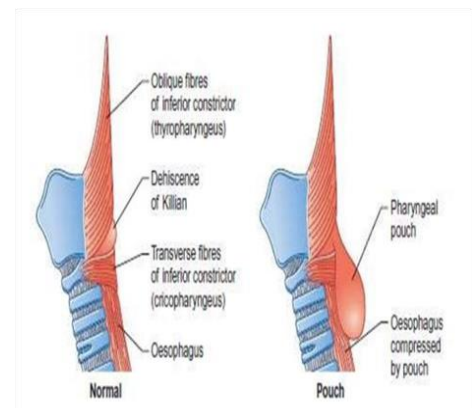


Figure 14: Pharyngeal pouch

Ludwig's Angina

- Rare, but serious connective tissue infection of the floor of the mouth
- Mostly due to dental infections. Signs of inflammation are present
- Treatment: drainage of pus + antibiotic to cover aerobes with anaerobes

Thyroid Masses

- Thyroid neoplasms are a common cause of anterior compartment neck masses in all age groups, with a female predominance, and are mostly benign.
- Fine needle aspiration of thyroid masses has become the standard of care and ultrasound may show whether the mass is cystic.
- Unsatisfactory aspirates should be repeated, and negative aspirates should be followed-up with a repeat FNAC and examination in 3 months' time.

MALIGNANT NECK LUMPS

Lymphoma

- Painless lump, non-tender, smooth, and discrete
- Slow-growing
- Patient presents with malaise, weight loss, pallor.
- Fever, rigor and hepato-splenomegaly
- Mediastinal mass (superior vena caval [SVC] syndrome)
- Abdomen pressure on the inferior vena cava (IVC) may cause bilateral leg edema
- Other LNs in the axilla, groin and abdomen should be examined.
- Treatment: according to stage (radio-sensitive)

Metastatic Lymph Nodes

- Upper cervical lymph nodes (upper aero-digestive tract).
- Accessory chain of nodes in the posterior triangle (Naso-pharyngeal malignancies).
- **Occult primary** most common sites are tonsil, base of tongue, naso-pharynx and pyriform sinus.
- Virchow's LN (**Troisier 's sign**) abdominal and thoracic malignancies
- Painless, non-tender, and hard masses
- Work-up: Search for primary and deal with it

CHAPTER

5

Surgery of the Salivary Glands

ANATOMY AND PHYSIOLOGY

Salivary glands are three paired large salivary glands (parotid, submandibular and sublingual) and hundreds of small salivary glands that open into the mouth.

The Parotid Gland

- It is the largest of the salivary glands. It overlies the ramus of the mandible and wraps around it medially to form superficial & deep lobes. It lies mainly anterior & inferior to the external auditory meatus (EAM), and also extends posteriorly to the area behind the lobe of the ear (Figure 1).

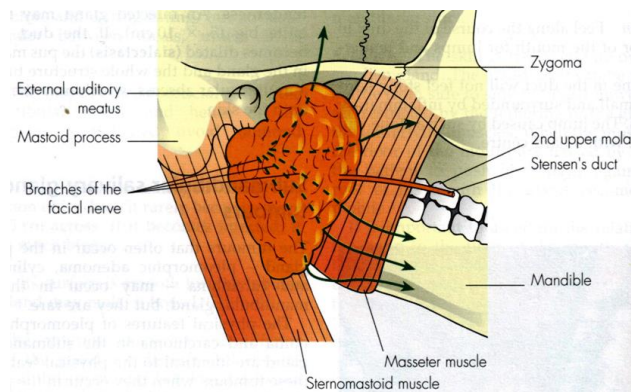


Figure 1. Anatomy of the parotid gland.

- *Function of the parotid gland:* The parotid gland is made up almost entirely of serous acini, which produce about 25% of the daily saliva. Stimulation of the gland is mainly via the parasympathetics of the lesser petrosal nerve, which is a branch of the glossopharyngeal nerve. Stimulation produces watery, amylase-rich saliva. On the other hand, sympathetic stimulation produces thicker glycoprotein-rich saliva. The main function is that it aids in food digestion and teeth protection.
- *Important relations in the parotid gland:*
 1. Three important structures pass through the gland from lateral to medial; namely, the facial nerve, retro-mandibular vein (drains to the internal jugular vein - IJV) & external carotid artery (ECA).
 2. Numerous lymph nodes (LNs) and *Stensen's duct* that opens into the mouth at the level of the upper 2nd molar tooth.

The Submandibular Gland

- The submandibular gland produces approximately 70% of the daily saliva. It has both serous and mucinous acini, but the serous predominate.
- It lies partly under cover of the body of the mandible and sits between the mandible and the hyoid bone. It also wraps around the mylohyoid, which separates it into superficial & deep lobes (Figure 2).
- *Superficial part:* It lies in the digastric triangle & is separated posteriorly from the parotid gland by the stylo-mandibular ligament. It is covered by the deep investing layer of cervical fascia, platysma & skin.
- *Deep part:* It extends forwards between the mylohyoid below and the hyoglossus and the styloglossus above.
- *Submandibular (Wharton's) duct:* It runs in the plane of the deep lobe anteriorly under the mucous membrane of the floor of the mouth. It opens into the anterior floor of the mouth just lateral to the frenulum of the tongue at a small papilla. Figure 3 shows the relation of the submandibular duct to the lingual nerve.

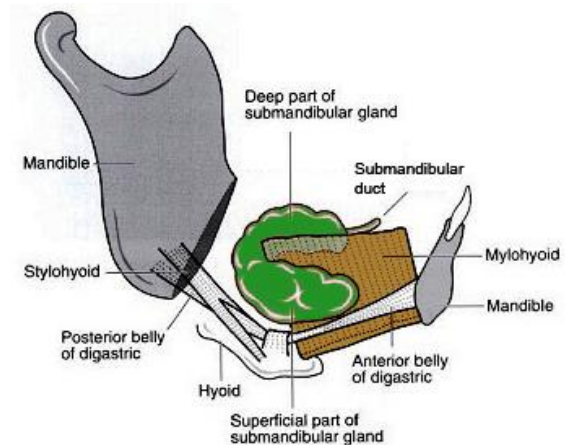


Figure 2. The U-shaped submandibular salivary gland around the mylohyoid muscle

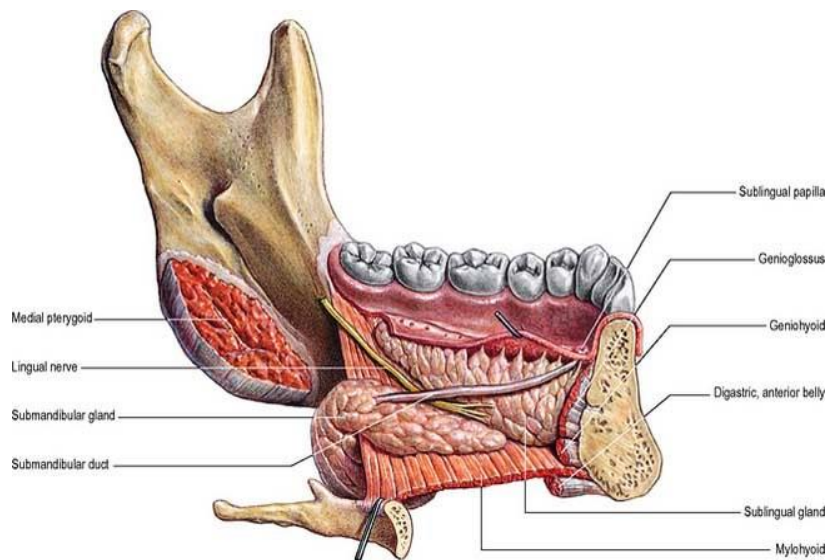


Figure 3. Relation of the lingual nerve to the submandibular duct

The Sublingual Gland

- The sublingual gland is the smallest of the three paired salivary glands. It lies beneath the mucous membrane of the floor of the mouth, either sides of the midline. It contains serous and mucinous.
- These glands should be noted for 2 main reasons: (1) *Tumors* of these glands are almost always malignant, and (2) A *ranula* is a blocked sublingual gland (mucocele) in the floor of the mouth (a plunging ranula is one that prolapses through the mylohyoid).

SALIVARY GLAND STONE DISEASE

History-Taking

The essential history of a patient with stones in the salivary gland (sialolithiasis) includes:

1. Acute painful swelling of one major salivary gland
2. Symptoms that are much worse after eating
3. Swelling reduces over 1-2 hours after eating
4. A stone may be felt in either Stensen's duct or Wharton's duct.
5. Swelling may be recurrent

Incidence

- Approximately, 80-90% occur in the submandibular gland and 10-20% in the parotid gland
- Can occur at any age
- Slight male predominance

Pathogenesis

- Risk factors: Dehydration, gout and hypertension.
- Most stones are made of hydroxyapatite (abundant in saliva). They form around a nidus of mineralized deposits within the salivary duct.
- As Wharton's duct is much longer than Stensen's duct, and the submandibular saliva has a higher content of mucin, calcium & phosphate, stones occurring in the submandibular duct are much common.

Imaging

- *Plain X-ray*: Due to the high content of calcium in the submandibular duct stones, 85% are radio-opaque and can be seen on a radiograph. Parotid saliva has a much lower concentration of calcium and only 10-15% of these are radio-opaque. Thus, not all stones are visible on a radiograph, so floor-of-the mouth and occlusal radiographs are no longer routinely used.
- *Ultrasonography* highlights most stones and gives anatomical details of the gland.

Submandibular Gland Surgery

- Important structures surrounding the gland that need to be taken into consideration when planning for excision include the facial artery and vein, the marginal mandibular nerve (MMN) (branch of the facial nerve), the lingual & hypoglossal nerves.
- *The MMN* runs on the deep surface of the platysma. It is variable in the distance where it drops below the body of mandible, so to avoid injury to this the following steps are taken:
 1. The incision is made 3-4 cm below the inferior margin of the mandible
 2. The platysma is incised at this low point
 3. The capsule of the gland is entered at its lowest point (for benign disease only)
- *The lingual nerve* is a branch of the mandibular nerve arising in the anatomy of the submandibular gland because it is found intimately associated with Wharton's duct and gives the parasympathetic supply to the gland. As it runs anteriorly, it first lies lateral, then inferior, then medial to Wharton's duct.

SALIVARY GLAND TUMORS

Salivary gland neoplasms can be divided into benign and malignant tumors (1^{ry} or 2^{ry} tumors) (Figure 4)

Incidence

- 80% of all salivary gland tumors are in the parotid gland
- 80% of parotid tumors are benign (80% of these are pleomorphic adenomas)
- 50% of tumors arising in the submandibular gland are benign. 10% of sublingual tumors are benign and <1% of minor salivary gland tumors are benign



Figure 4. A parotid gland tumor characteristically raises lobule of the ear

Classification

Adenoma

- Pleomorphic adenoma
- Adenolymphoma (Warthin's tumor)
- Myoepithelial
- Basal cell adenoma
- Ductal papilloma
- Cystadenoma

Carcinoma

- Mucoepidermoid carcinoma (most common malignancy in children and adults)
- Acinic cell carcinoma
- Adenoid cystic carcinoma
- Carcinoma ex-pleomorphic adenoma
- Undifferentiated carcinoma

Non-epithelial Tumors

- Hemangioma (most common tumor in children)
- Lymphangioma
- Lipoma

Lymphomas

Secondary tumors

Unclassified tumors

Tumor-like conditions: e.g. sialometaplasia, sialoadenitis).

Salivary Adenomas

Pleomorphic adenomas

- Account for 80% of benign parotid gland tumors.
- They occur most frequently in the 5th decade, equally in men & women.
- They have a pseudo-capsule and arise from myoepithelial cells and intercalated duct cells
- They present as a painless enlarging smooth mass
- Treatment is with excision because there is a 2% per year malignancy transformation rate.

Adenolymphomas (Warthin's tumor)

- Are seen mainly in men (7M:1F) aged 60-70 years
- They usually arise in the parotid tail from lymphoid tissue.
- One in 10 patients has bilateral tumors, but these rarely occur synchronously.
- They can be treated conservatively but, if cosmetic appearance is distressing, they can be removed

Salivary Carcinomas*Mucoepidermoid tumors (MET)*

- They arise mainly in the parotid gland.
- They are the most common salivary neoplasms in both children and adults
- These tumors metastasize to LNs and can spread to the lungs and brain.
- They may be of a low grade or a high grade (more aggressive)

Adenoid cystic carcinoma

- It is a slow-growing tumor that often spreads along the *nerve sheath*.
- It occurs more frequently in minor rather than major salivary glands.
- Patients may present with facial pain and facial nerve or trigeminal nerve palsy.
- Tumors do not metastasize early and LN metastasis is uncommon but, due to *perineural spread*, these cancers are often treated over many years and re-present many years later with new lesions.

Adenocarcinomas

- Make up 3% of parotid tumors and 10% of submandibular and minor salivary gland tumors, 20% of patients have nodal disease at presentation.

Carcinoma ex-pleomorphic adenoma

- It is a malignancy arising within a pre-existing benign pleomorphic adenoma.
- The incidence of malignant change is estimated to be 2% per year in an adenoma that has been present for >10 years.

Staging of Salivary Gland Neoplasms

The AJC system (American joint committee) Staging of malignant parotid tumors

T0	No clinical evidence of tumors
T1	<2cm diameter, without extra-parenchymal extension
T2	2-4 cm diameter, without extra-parenchymal extension
T3	4-6 cm diameter, and/or extra-parenchymal extension
T4a	Invasion of ear canal, skin, mandible of facial nerve
T4b	Base of skull, nerve VII involvement and/or >6 cm

N.B.

A parotid lump with facial nerve palsy is cancer until proved otherwise

Investigations

1. Ultrasonography (US)
2. MRI (especially dynamic MRI) helps to assess relation of tumors to anatomical structures (eg facial nerve) & helps to identify the type of the lesion.
3. FNAC

Parotid Gland Surgery

This aims to resect tumor with a margin of macroscopically normal tissue and to preserve the facial nerve.

- *Superficial parotidectomy*: It means excision of the superficial lobe only, but tumors extending to the deep lobe require *total parotidectomy* (deep and superficial lobes removed)
- *Total conservative parotidectomy* preserves the facial nerve.
- *Total radical parotidectomy* sacrifices the facial nerve
- *Extended parotidectomy* involves excision of additional structures (e.g. temporo-mandibular joint, mandible, zygoma or sternomastoid)
- *Facial nerve resection*: Adenoid cystic carcinomas may require parts of the facial nerve to be resected if they are infiltrated by tumor
- *Neck dissection* is indicated in nodal disease and may be indicated in high-grade tumors (eg a high-grade MET with an N0 neck will still get a neck dissection. A similarly sized low-grade tumor may *not* need a neck dissection)

Radiotherapy

- *Post-operative radiotherapy is indicated for:*
 1. Residual disease
 2. If there is evidence of extra-capsular spread in LNs
 3. High-grade tumors with high risk of local recurrence
 4. Surgery for recurrent disease
 5. Adenoid cystic tumors
 6. Perineural disease
- *Palliative radiotherapy* alone can be offered for inoperable cases.

CHAPTER

6

Oral Lesions and Jaw Swellings

LESIONS OF THE ORAL CAVITY

The oral cavity includes the mouth, floor of the mouth, cheek and tongue.

1. THE MOUTH

Stomatitis

Definition

- *Stomatitis* means inflammation of the mucous membranes of the oral cavity, while *gingivitis* means inflammation of the mucoperiosteum that covers the alveolar processes i.e. inflammation of the gum.

Causes

- Vitamin deficiencies: vitamin C (Scurvy) and vitamin. B12.
- Iron deficiency.
- Viral infections (Herpes, Cocksachie, HIV infection) and fungal infection.
- Mercury and lead poisoning.
- Corticosteroids and chemotherapy.

Aphthous Stomatitis

- Small, *multiple, painful* ulcers.
- Occur on non-keratinized mucosa of the oral cavity (lip, cheek, soft palate, floor of mouth)
- Associated with stress or decreased immunity.
- Normally heals within 10-14 days.
- May be associated with syndromes as Behcet's or Reiter's syndromes (Table 1 below).

Behcet's syndrome	Reiter's syndrome
- Multiple oral cavity ulcers - Conjunctivitis - Genital lesions	- Multiple oral cavity ulcers - Conjunctivitis - Urethritis. - Arthritis - Polyarteritis nodosa (PAN)

Angular Stomatitis

- *Definition:* It is infection and fissuring at the corners of the mouth.
- *Etiology:* It is caused by continuous leak of saliva at the angles leading to a moist skin which becomes infected by candida and staphylococci.
- *Age:* This occurs in old patients (edentulous) or babies who suck their thumb.

2. FLOOR OF THE MOUTH

Salivary Mucous Cysts

1. Mucous cysts

- Arise from minor salivary glands.
- Appear on the inner surface of the lips or the cheek.
- Present as pinkish, bluish, soft cysts.
- Treated by excision.

2. Ranula

- The Latin word *rana* means "frog" (ranula = little frog), the term is usually explained by a resemblance with the bulging throat of the *frog*.
- It is a bluish translucent cyst of the sublingual salivary gland (Figure 1).
- If it grows bigger, it will cross the posterior border of mylohyoid muscle to appear in the neck & is called *plunging* ranula.
- Treatment:
 - a) Treatment is by *excision* of the cyst with the gland intra-orally. If it is plunging, treatment is also excision intra-orally with insertion of a drain from the floor of the mouth down into the neck cavity.
 - b) If complete excision is *not* possible, *marsupialization* is performed.



Figure 1. Ranula. Bluish translucent cyst

Lingual and sublingual Dermoid Cyst

- It is lined by squamous cells (producing keratin) or mucous-secreting cells (producing mucous).
- It usually presents at the midline.
- It is treated by excision through a vertical incision from tip of tongue to the frenum.

3. THE CHEEK

Cheek Swellings

- *Benign:* Papilloma (cauliflower mass), hemangioma, denture-induced granuloma, lipoma, neurofibroma.
- *Malignant:* Carcinoma

Cheek Squamous Cell Carcinoma (SCC)

- It occurs in heavy smokers, spirit drinkers and betel nut chewers. It also occurs on top of leukoplakia.
- Treatment
 - Wide local excision + LN neck dissection + post-operative radiotherapy (sine qua non)
 - Reconstruction of the buccal mucosa:
 - a) Split thickness skin graft.
 - b) Pedicled flap (delto pectoral or forehead flap).
 - c) Free flap e.g. sandwich TRAM, or radial forearm flap (Figure 2).

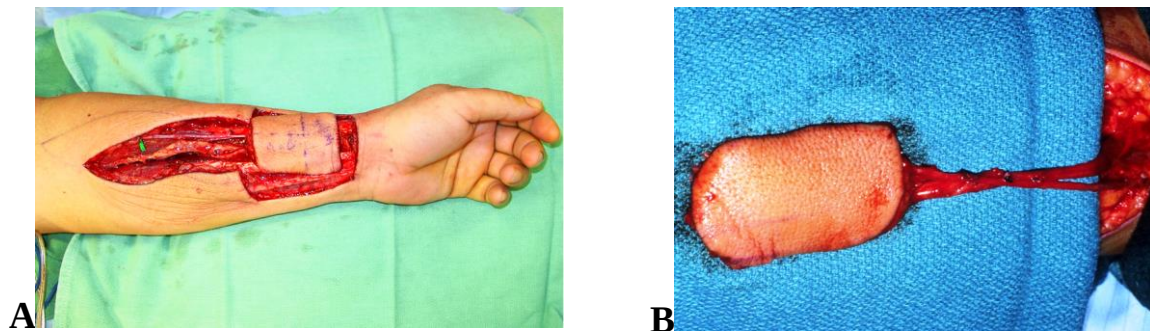


Figure 2. Elevation of radial forearm flap (A), flap elevated & ready for transfer (B)

4. THE TONGUE

A. Congenital (Developmental) Anomalies

- Tongue-tie.
- Congenital bifid tongue.
- Lingual thyroid: reddish lobulated mass behind the foramen cecum.
- Macroglossia: Neurofibromatosis, cavernous hemangiomas, lymphangioma.

B. Hyperkeratosis & Leukoplakia

- Both terms more or less have the same meaning.
- Appear as hyper-keratotic, *white* patches (Fig.3).
- Hyperkeratosis reverses back to normal once the offending agent is stopped e.g. smoking & friction. Leukoplakia, however, does *not* revert back & the offending agent is not usually known.
- *Causes:* Smoking, spices, sepsis, spirits, sharp teeth, syphilis, candida



Figure 3. Leukoplakia. Note the white patches

C. Tongue Ulcers

Traumatic	Dental
Inflammatory	- Viral: Cocksackie, Herpes - Bacterial: Acute (Pertussis), Chronic (TB [tip] AND syphilis [middle]). - Fungal
Dyspeptic	Aphthous
Neoplastic	SCC

D. Tongue Cancer

Risk Factors Associated with Cancer of the Head and Neck

- Tobacco, alcohol, Areca nut/pan masala, Human papilloma virus, Epstein–Barr virus, Poor nutrition and Sideropenic dysphagia (*Plummer-Vincent* and *Paterson-Kelly* syndromes)
- *Sideropenic dysphagia*: There is a well-known relationship between sideropenia (iron deficiency in the absence of anemia) & the development of oral cancer. Sideropenia is common in Scandinavian women and leads to epithelial atrophy, which renders the oral mucosa vulnerable to irritation from topical carcinogens. Correction of the sideropenia with iron supplements reduces the epithelial atrophy and risk of malignant transformation.

Incidence and Site

- *Gender*: Males > females in a ratio 2:1 or 3:1
- *Mostly* on the lateral margin (47 %) (Figure 4)
- Ventral surface (9 %)
- Dorsum (6 %) - *Never* in the midline
- Posterior 1/3 (20 %)

Etiology

- *Risk factors*
Smoking, spices, spirits, sharp teeth, sepsis, syphilis, candida.
- *Premalignant lesions*
Leukoplakia, erythroplakia, oral submucous fibrosis.

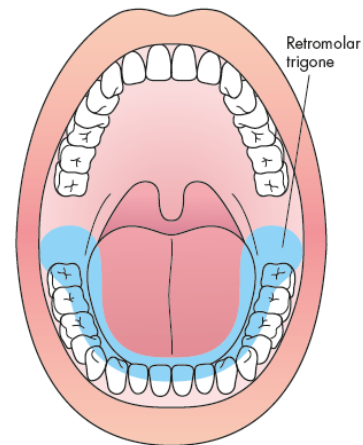


Figure 4. Common anatomical sites (blue) of oral SCC

Pathology

- *Microscopically*: SCC (most common) – sarcoma – malignant melanoma
- *Grossly*: Ulcer (most common) – diffuse infiltrative (woody tongue) – cauliflower mass

Clinical picture

- Malignant ulcer: Everted edges are characteristic (Figure 5)
- Raised lobulated hard mass (verrucous carcinoma)
- Raised plaque
- Pain in the tongue or referred to the ear due to involvement of the lingual nerve.
- Excessive salivation
- Ankyloglossia.
- Dysphagia
- Inability to articulate well
- Feter
- Lump in the neck



Figure 5. Tongue cancer (SCC)

Investigations

- *Biopsy*: Excisional (for small lesion under local anesthesia) - incisional (for a large lesion)
- *FNA* for the neck nodes (reactive or malignant)
- *Examination under anesthesia* (EUA)
- *Imaging*: Ultrasound (US) to the neck for LN status - CT scan - MRI
- *Endoscopy*: laryngoscopy, pharyngoscopy, or naso-pharyngoscopy

Classification and Staging

- Staging of head and neck cancer is defined by the American Joint Committee on Cancer (AJCC) and follows the *TNM system* (Tables 2 and 3).

Table 2: TNM Classification

Tx	Primary tumor cannot be assessed
T0	No evidence of primary tumor
Tis	Carcinoma in-situ.
T1	Tumor <2 cm (in greatest dimension).
T2	Tumor >2 but <4 cm
T3	Tumor >4 but <6 cm.
T4	Tumor invades adjacent structures (e.g. mandible, skin).
Nx	Regional LNs cannot be assessed
N0	No regional LN metastases.
N1	Metastasis in a single ipsilateral LN <3 cm (in greatest dimension)
N2a	Metastasis in a single ipsilateral LN >3 cm but not > 6 cm.
N2b	Metastasis in multiple ipsilateral LN, none > 6 cm.
N2c	Metastasis in bilateral or contralateral LNs, none > 6 cm.
N3	Metastasis in any LN>6 cm
M0	No evidence of distant metastases.
M1	Evidence of distant metastases.

Table 3: Stage Grouping

Stage	Tumor (T)	Nodes (N)	Metastases (M)
0	Tis	N0	M0
I	T1	N0	M0
II	T2	N0	M0
III	T3	N0	M0
	T1,T2,T3	N1	M0
IV	Any T	N2	M0
	Any T	N3	M0
	Any T	Any N	M1

Treatment

- Wide local excision with a 2-cm safety margin.
- If LNs are +ve: Unilateral Modified Radical Neck Dissection (MRND)
- If LNs are -ve: *selective* supra-omohyoid neck dissection (on same side of tumor)
- If SCC at the tip and +ve LNs: Bilateral MRND
- Post-operative radiotherapy (RT) (*sine qua non*)
- Reconstruction: Radial forearm free flap or pedicled-flap (pectoralis major or forehead)

Treatment of oropharyngeal cancer

- Tumors of the oropharynx are frequently not amenable to surgery because of the morbid nature of the resection.
- Tumors of the soft palate and tonsil, however, can be managed with either 1st surgery in continuity with neck dissection or 1st RT.
- Chemo-radiotherapy is now increasingly used to manage tumors of the oropharynx in which organ preservation, but not necessarily function, is the goal. In patients with large-volume neck disease, e.g. N2 and N3, a combined modality of neck dissection followed by chemo-radiotherapy to manage the 1st tumor and residual neck disease is gaining popularity

SWELLINGS OF THE JAW

Classification

1. Congenital
2. Traumatic
3. Inflammatory
4. Neoplastic
5. Others (cysts)

1. CONGENITAL

Odontomes (developmental abnormalities of the teeth)

A. Ameloblastoma (Adamantinoma)

Origin: it develops from dental lamina. It is the commonest tumor of the lower jaw.

Clinical picture

- It presents at the 4th or 5th decade of life, more common in females.
- It occurs at the 3rd molar region & may extend into ramus, angle, coronoid process or body.
- It may vary from soft to solid mass, which is painless, unless it is infected.
- It may present as a single cyst or a *multilocular* cyst, which is the commonest presentation.
- It is locally malignant.

X-ray

- Honeycomb or fine soap & bubble appearance (Figures 1 and 2)

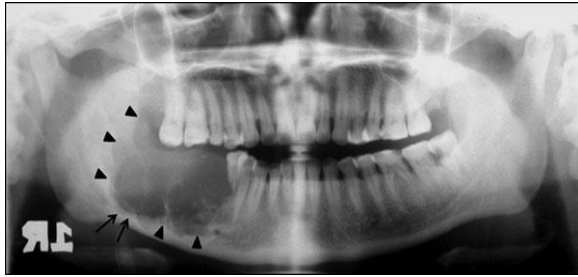


Figure 1



Figure 2

Figures 1 and 2. Plain X-ray of the mandible showing the soap and bubble appearance of ameloblastoma

Treatment

- Excision with 1 cm safety margin with reconstruction using bone grafts.
- If not adequately removed, it will give rise to lung metastasis (no lymph node metastasis).
- It is *not* radio-sensitive (i.e. it is *radio-resistant*).

B. Odontogenic Myxoma

- Similar to ameloblastoma, but it has myxomatous stroma.

2. TRAUMATIC

- Mandible fractures are a frequent injury because of the mandible's prominence AND relative lack of support.
- Most fractures occur in the body (29%), condyle (26%) AND angle (25%) of the mandible. The symphysis accounts for 17%, whereas fractures of the ramus (4%) and coronoid process (1%) have lower occurrence rates (Figure 3).
- Mandibular fractures present with an overlying hematoma and tenderness. Inability to open the mouth occurs in fracture condyle.
- A panoramic view gives a good picture of the mandible, teeth and their roots.
- Chest X-Ray may also be required if teeth are unaccounted for.

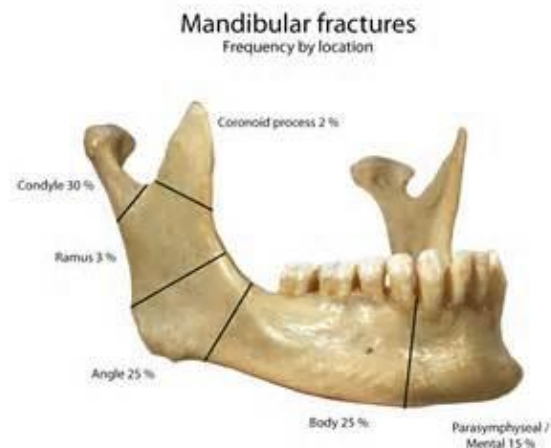


Figure 3. Sites of mandibular fractures

3. INFLAMMATORY

A. Alveolar Abscess

- *Definition:* It is an abscess in the alveolar ridge of the jaw, usually caused by the spread of infection from an adjacent non-vital tooth. It is also called dental abscess.
- *Clinically* there is throbbing pain and overlying tenderness, redness, hotness and swelling.
- *Treatment:* Successful treatment of a dental abscess centers on the reduction and elimination of the offending organisms. This includes treatment with antibiotics and drainage. If the tooth can be restored, root canal therapy can be performed. Non-restorable teeth must be extracted.

B. Osteomyelitis of the Mandible

- It may be acute or chronic.
- It usually follows extraction of a tooth from an irradiated mandible for treatment of squamous cell carcinoma.
- Clinically: swelling, trismus and draining sinus.
- Diagnosis is by Panorax X-ray.
- Treatment: Sequestrectomy.

C. Cervico- Facial Actinomycosis

- Nodular, tender, warm and fixed to skin, with multiple sinuses and sulfur granules.
- *Treatment:* it is sensitive to penicillin.

4. NEOPLASTIC

Classification

A. *Benign:* Epulides - giant cell granuloma - aneurysmal bone cyst - hemangioma (endosteal)

B. *Malignant*

- Primary: SCC from overlying mucosa - osteoclastoma (locally malignant).
- Secondary: Lymphoma (Burkitt's) – myeloma – histiocytosis X.

Epulides

Definition

These are swellings (sessile or pedunculated mass of granulation tissue in front of the teeth) arising from the muco-periosteum (lumps on gums).

Types

1. Fibrous epulides

- The commonest form & usually lies between the incisor teeth of the lower jaw.
- The swelling is sessile at the beginning & then becomes pedunculated (Figure 4).
- Treatment: Surgical excision.



Figure 4. Fibrous epulides between incisor teeth of the lower jaw

2. *Giant cell epulides*

- Arise from the osteoblastic layer of the periosteum.
- It contains osteoclasts and is considered an *osteoclastoma of the alveolar margin*.
- The swelling is most commonly situated peripherally in the mandible.
- It is rapidly growing and liable to necrosis, ulceration, cyst formation and malignant transformation.
- Treatment: Surgical excision.

3. *Pregnancy epulides*

- It results from increased estrogen level during pregnancy and bad oral hygiene.
- The patient presents with hypertrophy of the gums and a tendency to bleed.
- It is treated by good oral hygiene and avoidance of trauma.

4. *Denture induced granuloma*

- It is a reactive lesion to chronic mechanical irritation produced by a poorly fitting denture.
- It is a harmless condition & does not represent oral cancer.
- Treatment is by simple surgical removal of the lesion and also by adjustment of the denture or provision of a new denture.

5. *Pyogenic granuloma*

- Pyogenic granulomas usually present as smooth or lobulated masses that may be either pedunculated or sessile.
- The treatment of choice is surgical excision.

Osteoclastoma (Giant Cell Tumor)

- It is a rare locally malignant tumor of the jaw.
- It has a central predilection (i.e. arises near the symphysis menti).
- It has a resemblance to giant cell granuloma, giant cell epulides, ameloblastoma and brown tumor of hyperparathyroidism.
- Complete surgical resection is required for adequate treatment as incomplete resection leads to recurrence in about 70 % of cases (Figure 5)
- It is *radiosensitive*.

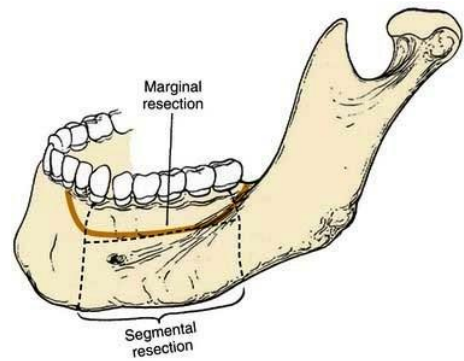


Figure 5. Marginal resection (yellow line) and segmental resection (dotted line) of the mandible.

Squamous Cell Carcinoma (SCC)

- It arises from the overlying mucosa or from adjacent floor of the mouth.
- Metastasizes to the deep cervical LNs and adjacent tissues and also through dental canal.
- *Treatment:* segmental resection of the mandible (Figure 5) together with adjacent involved tissues + modified radical neck dissection + post-operative radiotherapy.

Burkitt's Lymphoma

- *Etiology:* Epstein-Barr (EB) virus is the most likely etiological factor.
- *Age:* It occurs more commonly in children.
- *Clinically:* it most commonly presents by jaw swelling (osteolytic lesion of the mandible).
- *Diagnosis:* Plain X-ray or CT will show multiple osteolytic deposits with bone destruction.
- *Treatment:* the tumor is sensitive to chemotherapy.

5. OTHERS (CYSTS)

Dentigerous Cyst

- It arises from the crown of a non-erupted tooth.
- The crown protrudes through the cyst cavity (Figure 6).
- *Treatment*: enucleation.



Figure 6. PXR showing a dentigerous cyst:
Note the non-erupted tooth.

Dental Cyst

- Similar to dentigerous, but arises from the apex of the tooth (Figure 7).
- *Treatment*: enucleation.



Figure 7. PXR showing a dental cyst
arising at the apex of the tooth.